

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because $\text{np.isnan} > 15\%$: ['r_NH3_fft_3h_fft_f2_cph', 'r_NH3_fft_3h_fft_period2_h', 'r_NH3_fft_3h_fft_power2', 'r_NH3_fft_3h_fft_f3_cph', 'r_NH3_fft_3h_fft_period3_h', 'r_NH3_fft_3h_fft_power3', 'r_temperature_fft_3h_fft_f2_cph', 'r_temperature_fft_3h_fft_period2_h', 'r_temperature_fft_3h_fft_power2', 'r_temperature_fft_3h_fft_f3_cph', 'r_temperature_fft_3h_fft_period3_h', 'r_temperature_fft_3h_fft_power3', 'r_umidity_fft_3h_fft_f2_cph', 'r_umidity_fft_3h_fft_period2_h', 'r_umidity_fft_3h_fft_power2', 'r_umidity_fft_3h_fft_f3_cph', 'r_umidity_fft_3h_fft_period3_h', 'r_umidity_fft_3h_fft_power3']

1.2 Metadata

Nombre de lignes df	8052
Nombre de lignes df clean	8052
Nombre de features	14
% de NaN	0.0
Taille dataset entraînement	5636
Taille dataset test	2415
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_3h_fft_f1_cph, r_NH3_fft_3h_fft_period1_h, r_NH3_fft_3h_fft_power1, r_temperature_fft_3h_fft_f1_cph, r_temperature_fft_3h_fft_period1_h, r_temperature_fft_3h_fft_power1, r_umidity_fft_3h_fft_f1_cph, r_umidity_fft_3h_fft_period1_h, r_umidity_fft_3h_fft_power1

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

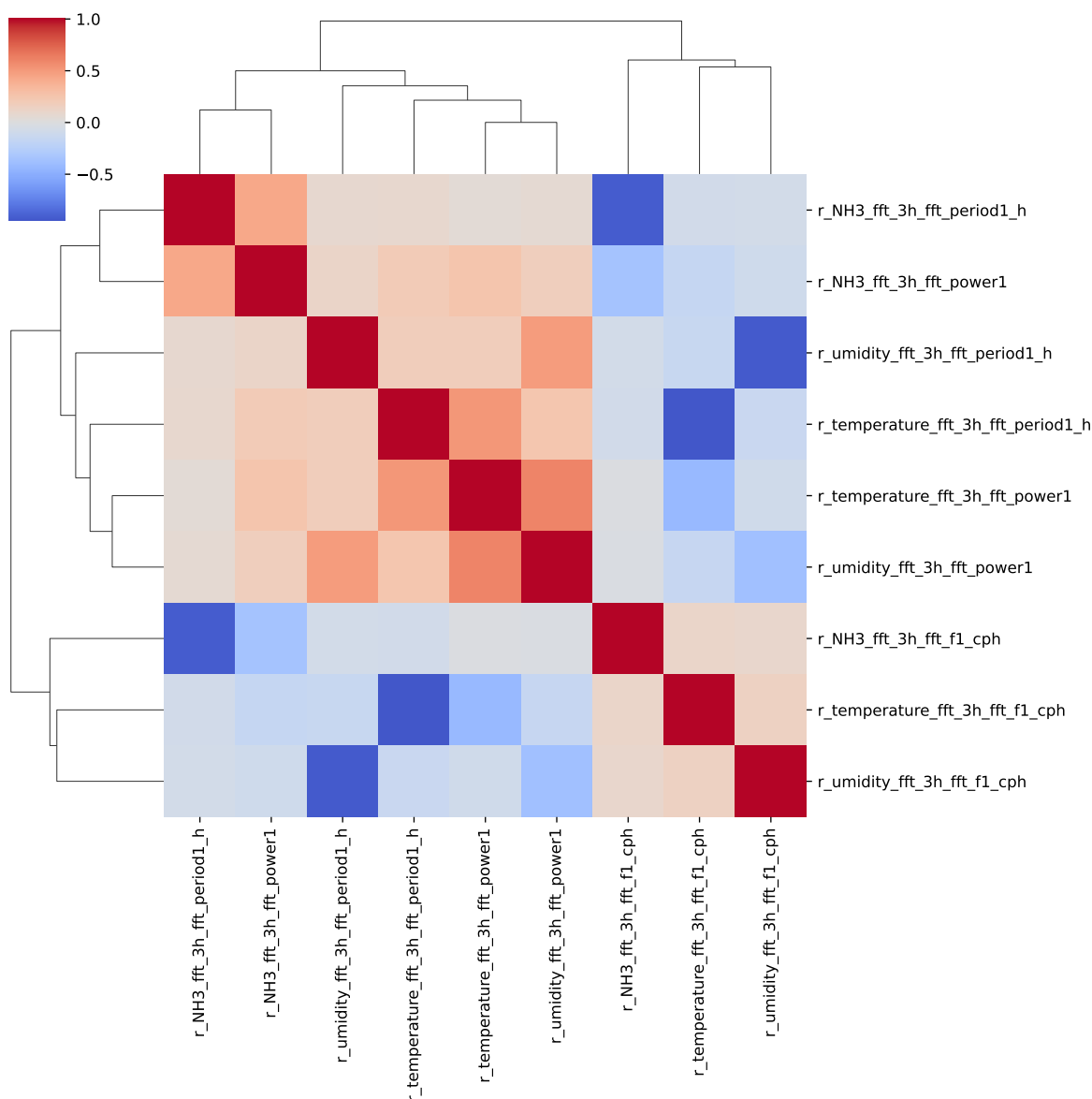
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_fft_3h_fft_f1_cph, r_NH3_fft_3h_fft_period1_h ; **cluster 7:** r_NH3_fft_3h_fft_power1 ; **cluster 8:** r_temperature_fft_3h_fft_f1_cph, r_temperature_fft_3h_fft_period1_h ; **cluster 9:** r_temperature_fft_3h_fft_power1 ; **cluster 10:** r_umidity_fft_3h_fft_f1_cph, r_umidity_fft_3h_fft_period1_h ; **cluster 11:** r_umidity_fft_3h_fft_power1 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_3h_fft_f1_cph, r_NH3_fft_3h_fft_power1, r_temperature_fft_3h_fft_f1_cph, r_temperature_fft_3h_fft_power1, r_umidity_fft_3h_fft_f1_cph, r_umidity_fft_3h_fft_power1



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_3h_fft_f1_cph, r_NH3_fft_3h_fft_power1, r_temperature_fft_3h_fft_f1_cph, r_temperature_fft_3h_fft_power1, r_umidity_fft_3h_fft_f1_cph, r_umidity_fft_3h_fft_power1

Len(Features_X): 11

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

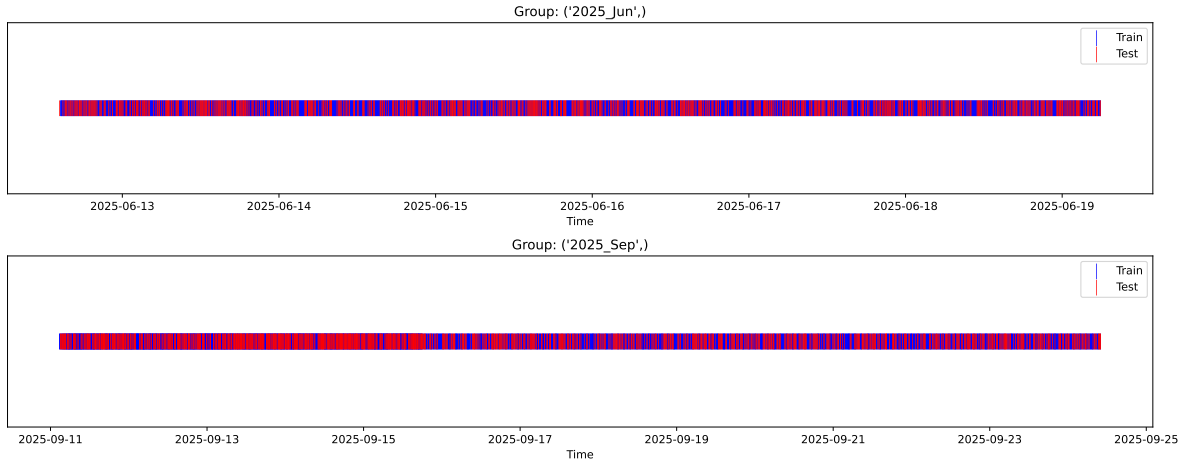
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

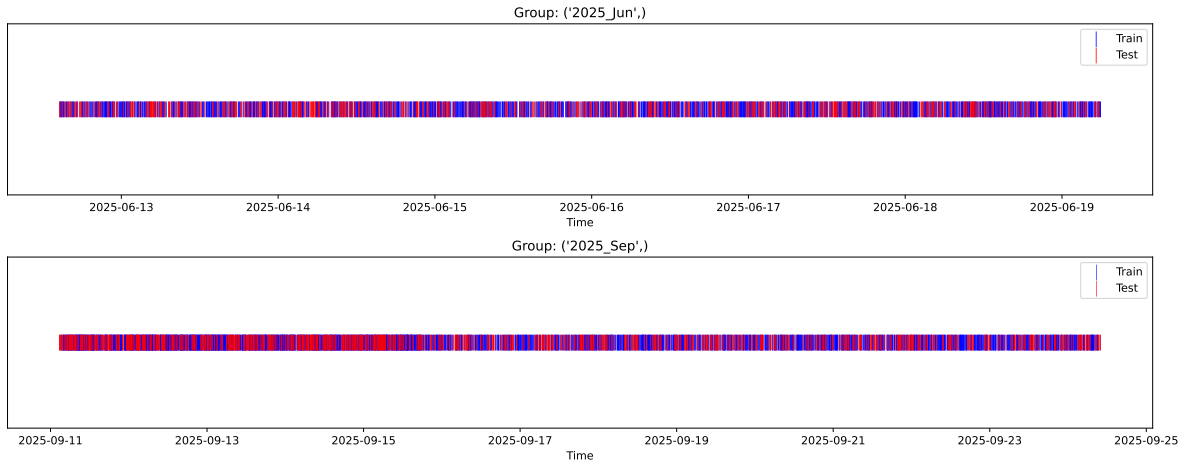
External Split: Train: 5636 samples Test: 2416 samples



3.3.2 Internal CV Folds

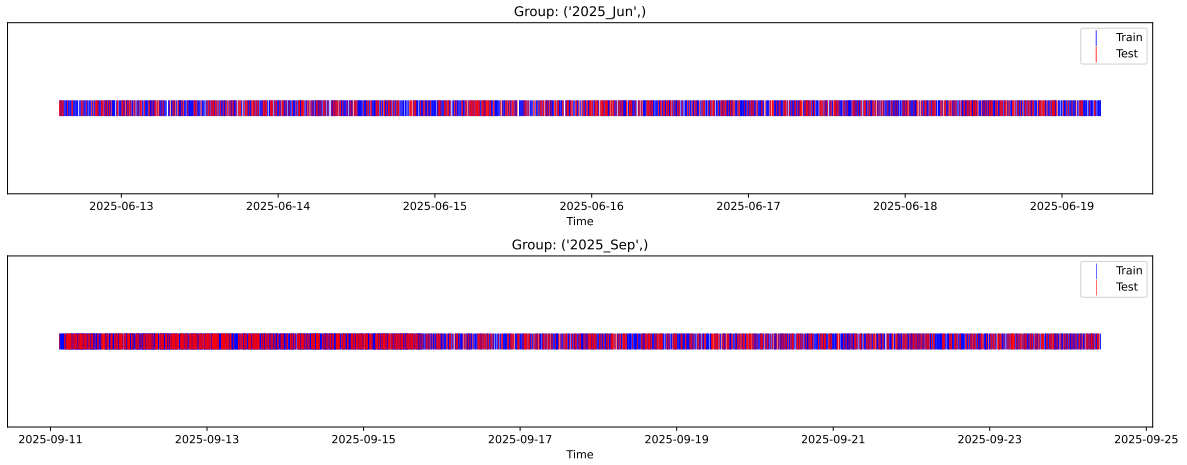
Fold 1/5

Train: 3945 samples Test: 1691 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3945 samples Test: 1691 samples Excluded: 0 samples Window exclusion: 0h



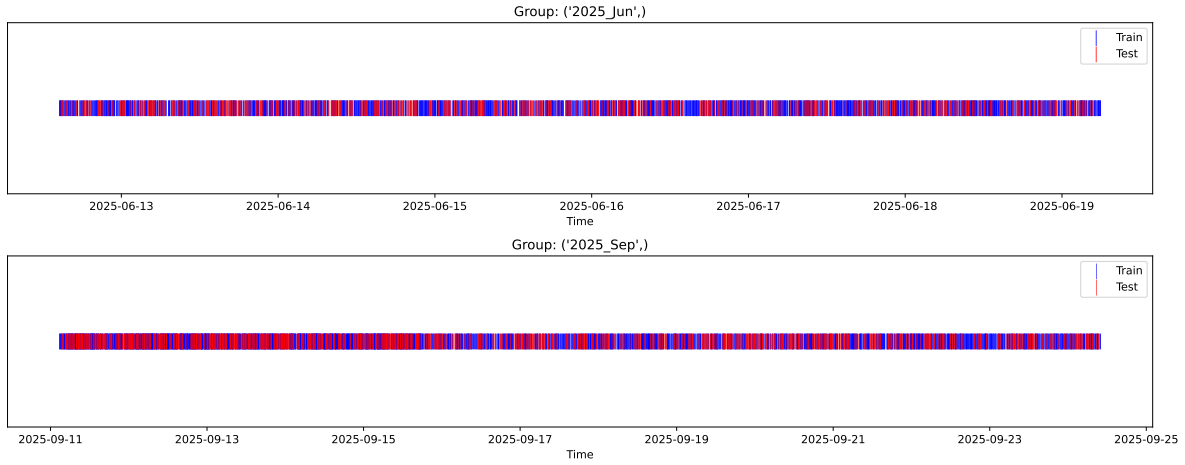
Fold 3/5

Train: 3945 samples Test: 1691 samples Excluded: 0 samples Window exclusion: 0h



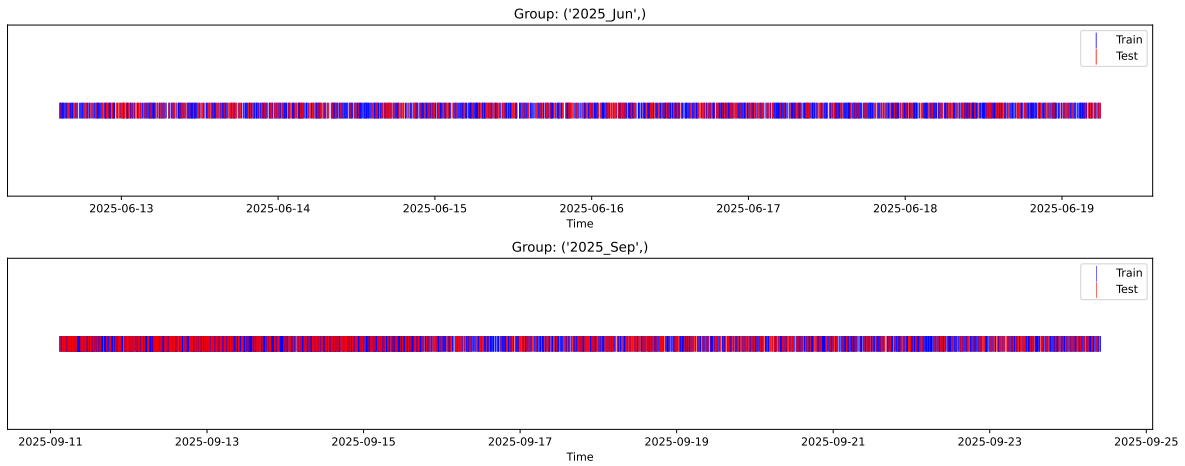
Fold 4/5

Train: 3945 samples Test: 1691 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3945 samples Test: 1691 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.804, min=0.959, max=13.845
 y_{test} : mean=2.369, std=0.782, min=0.973, max=7.822

Fold 0 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.81

Fold 1 — y_{test} : mean=2.36, std=0.84, y_{train} : mean=2.34, std=0.79

Fold 2 — y_{test} : mean=2.35, std=0.81, y_{train} : mean=2.34, std=0.80

Fold 3 — y_test: mean=2.33, std=0.79, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.33, std=0.76, y_train: mean=2.35, std=0.82

train scores CV: [0.44538672 0.47465262 0.46501286 0.44749598 0.45075338]

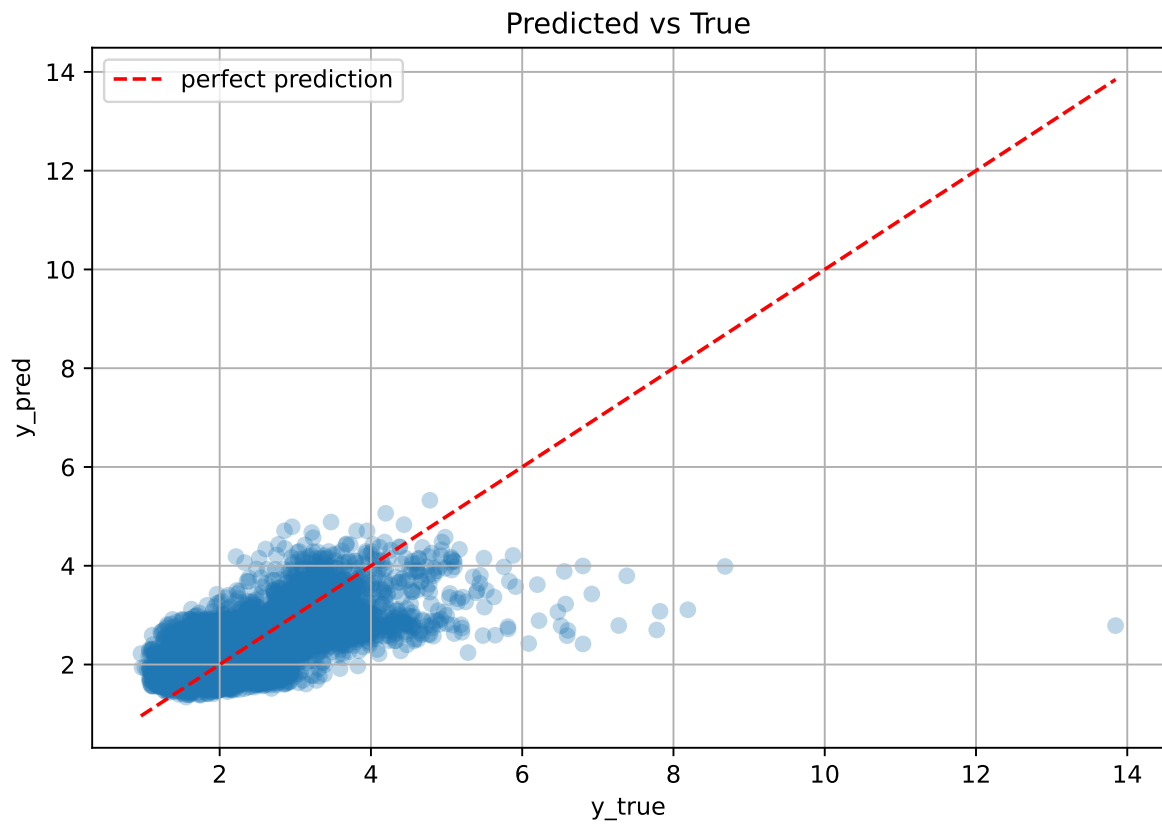
test scores CV: [0.47998273 0.41548957 0.43383685 0.47609704 0.46936504]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4567	0.455	0.0256	0.4621	0.4627	0.0078	1.0171	1.0037

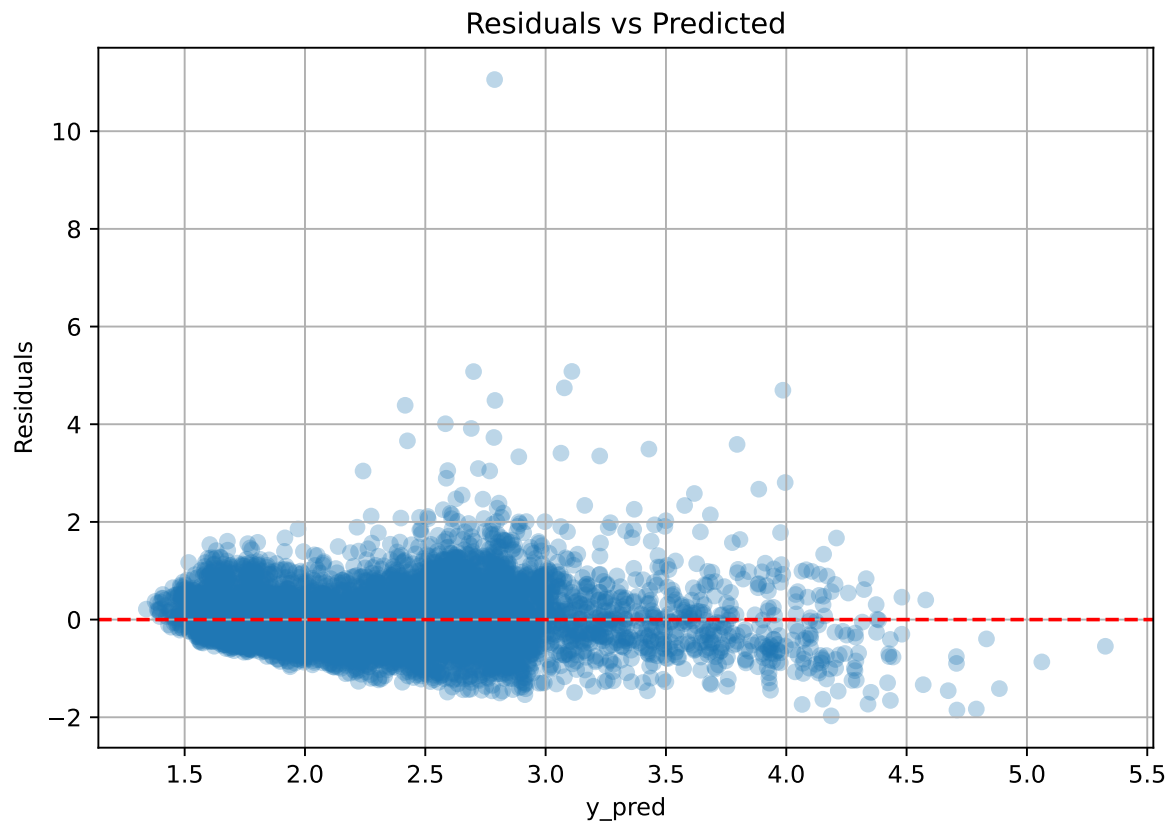
Coefficients:

coef_r_CO2 = -0.0406, coef_r_NH3 = -0.0732, coef_r_NH3_fft_3h_fft_f1_cph = 0.0164, coef_r_NH3_fft_3h_fft_power1 = 0.0258, coef_r_THI = 0.3509, coef_r_temperature = -0.3939, coef_r_temperature_fft_3h_fft_f1_cph = 0.0488, coef_r_temperature_fft_3h_fft_power1 = 0.2459, coef_r_umidity = -0.4583, coef_r_umidity_fft_3h_fft_f1_cph = 0.0111, coef_r_umidity_fft_3h_fft_power1 = 0.0611, intercept = 2.3438,

3.4.2 Scatter



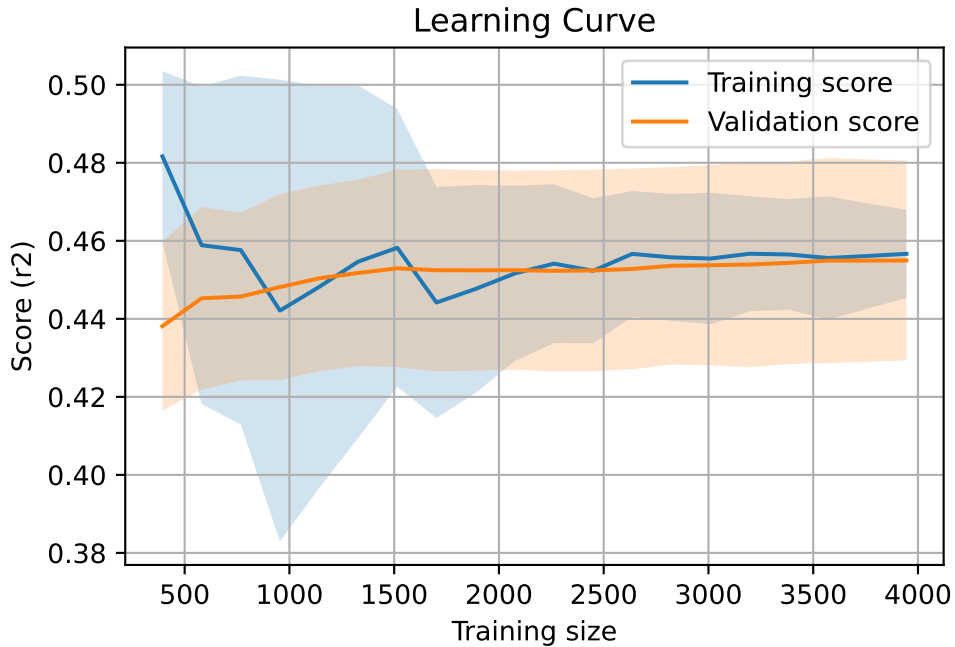
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 8052, `Len(training)` : 5637,

`Len(training_real)` : 3946, `Len(test_of_training)` : 1691, `Len(test)` : 2415



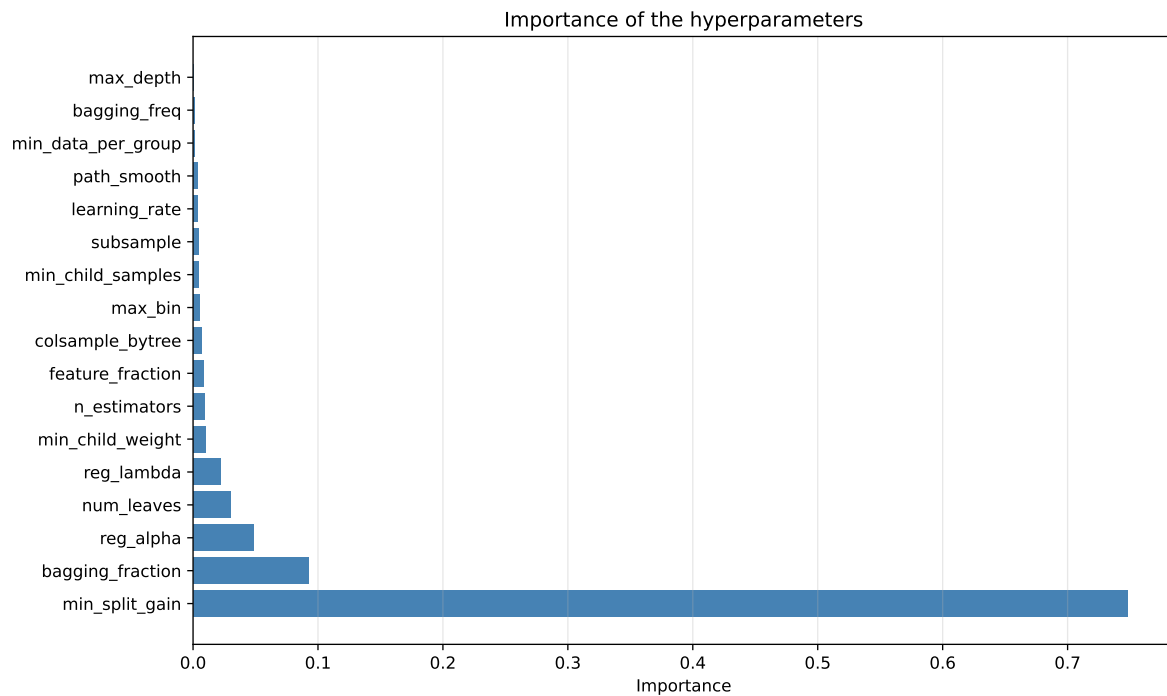
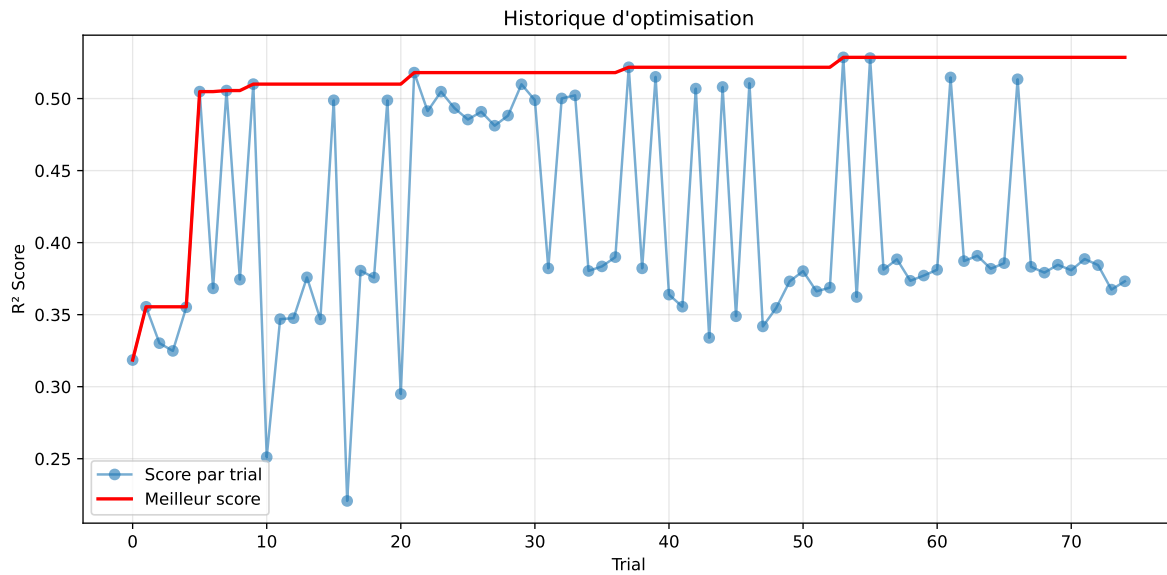
3.5 Model : LGBM

3.5.1 Gridsearch

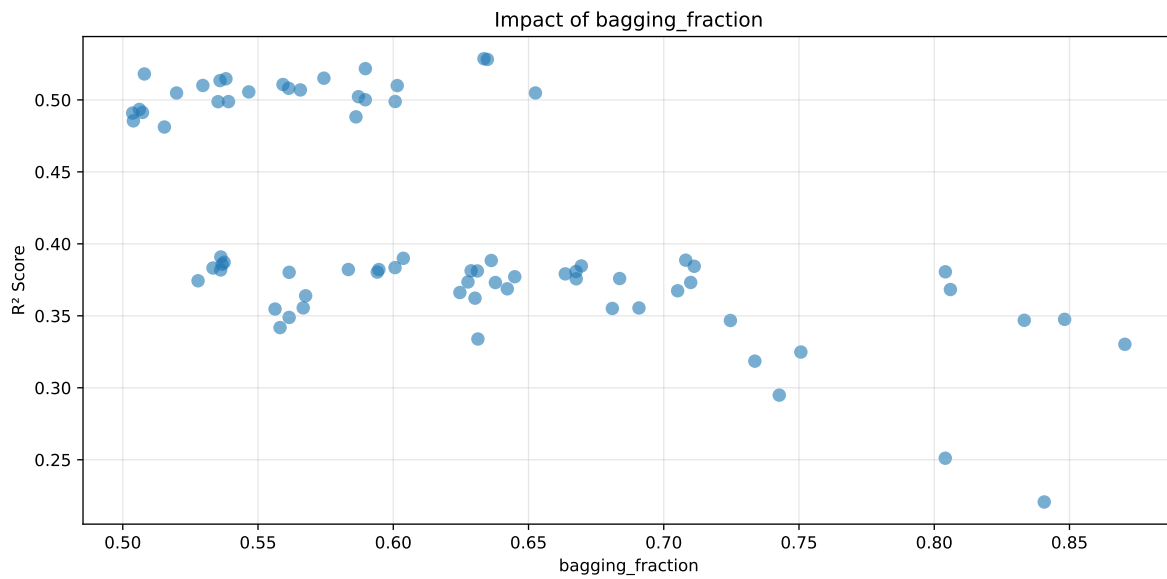
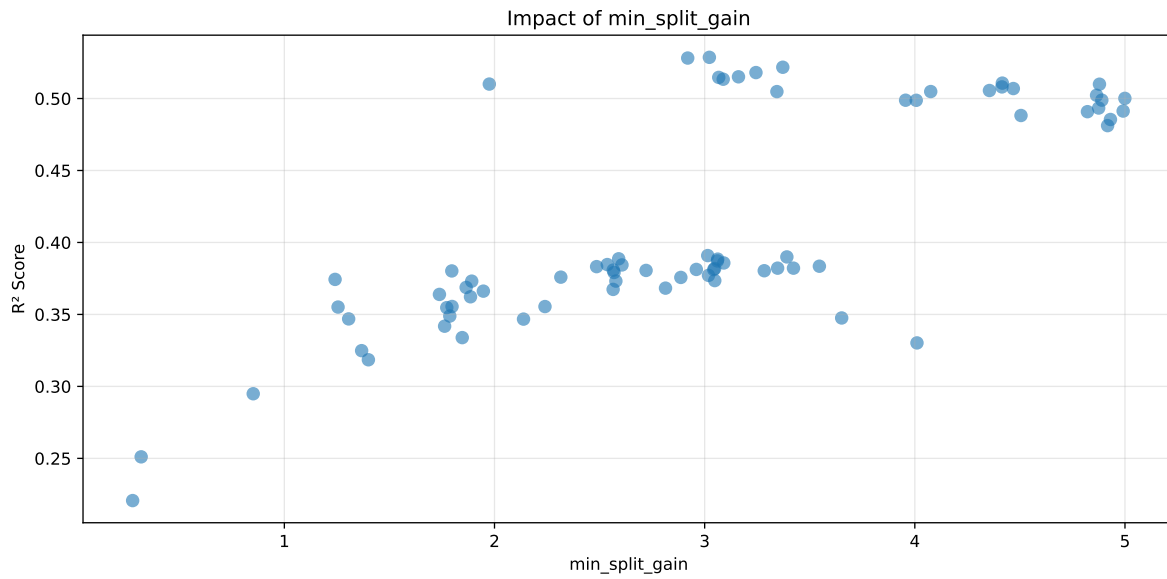
Distribution y_train vs y_test: y_train: mean=2.344, std=0.804, min=0.959, max=13.845
y_test: mean=2.369, std=0.782, min=0.973, max=7.822

===== TOP Importance FEATURES =====

feature	importance
r_CO2	104
r_umidity	68
r_temperature_fft_3h_fft_power1	62
r_NH3	32
r_THI	30
r_NH3_fft_3h_fft_power1	28
r_temperature	18
r_umidity_fft_3h_fft_power1	14
r_temperature_fft_3h_fft_f1_cph	10
r_NH3_fft_3h_fft_f1_cph	3
r_umidity_fft_3h_fft_f1_cph	3

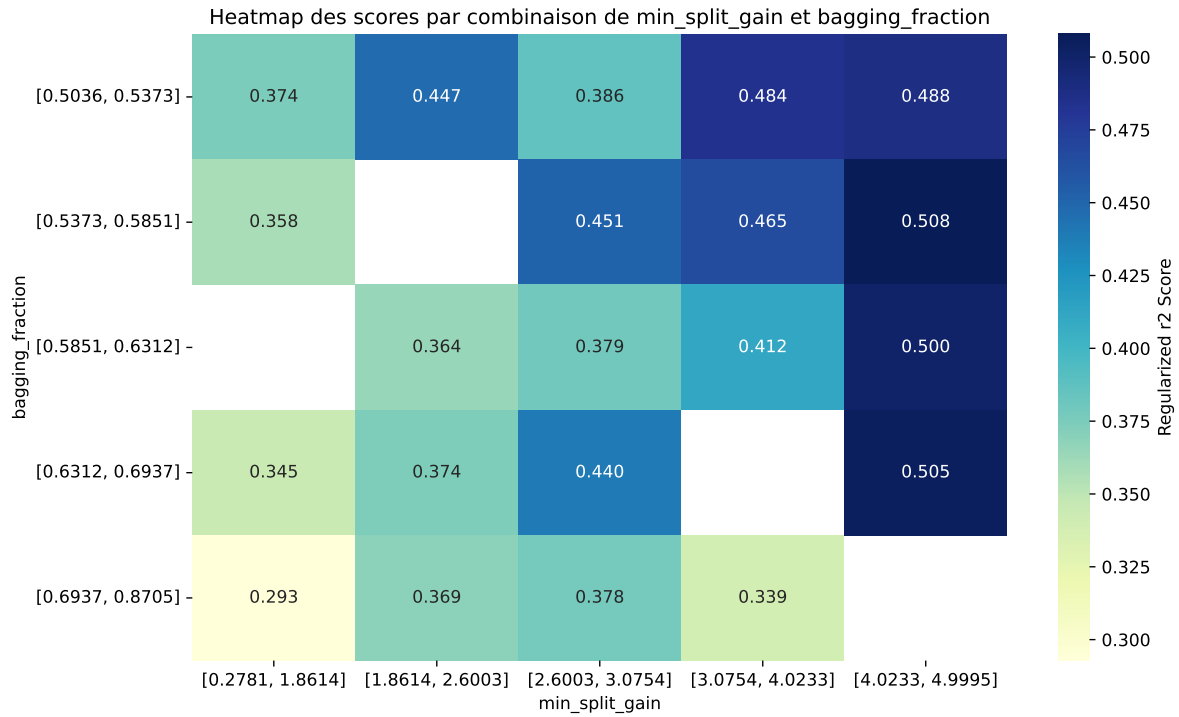


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5529	0.5286	0.5312	0.0206	0.5286	0.5465	0.0179	1.0339	1.046
2	0.5534	0.5281	0.5319	0.0221	0.5281	0.5431	0.015	1.0284	1.0478
3	0.5469	0.5217	0.5268	0.0231	0.5217	0.5505	0.0288	1.0553	1.0483
4	0.5416	0.518	0.5248	0.0263	0.518	0.5425	0.0245	1.0473	1.0456
5	0.5376	0.5151	0.5219	0.0206	0.5151	0.5436	0.0286	1.0555	1.0438
6	0.5389	0.5147	0.5233	0.0201	0.5147	0.5539	0.0392	1.0762	1.0471
7	0.5366	0.5134	0.5196	0.0203	0.5134	0.5377	0.0243	1.0473	1.0453
8	0.5338	0.5107	0.5145	0.0186	0.5107	0.5277	0.017	1.0333	1.0453
9	0.5336	0.51	0.5164	0.0223	0.51	0.534	0.0239	1.0469	1.0462
10	0.5344	0.5099	0.5135	0.0205	0.5099	0.5471	0.0372	1.073	1.0479
---	---	---	---	---	---	---	---	---	---
75	0.6794	0.6029	0.6173	0.0213	0.2207	0.6364	0.0334	1.0554	1.1268

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.07446405846817832, 'reg__num_leaves': 22, 'reg__subsample': 0.8461453613839403, 'reg__colsample_bytree': 0.705185623714098, 'reg__min_child_samples': 20, 'reg__reg_alpha': 4.258877447168149, 'reg__reg_lambda': 37.864457675002264, 'reg__min_split_gain': 3.0224314411480675, 'reg__path_smooth': 1.6696968600913498, 'reg__min_child_weight': 3.841226233805145, 'reg__feature_fraction': 0.8892997444585451, 'reg__bagging_fraction':

0.6335144019614446, 'reg__bagging_freq': 3, 'reg__max_bin': 97, 'reg__min_data_per_group': 93}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.0732692328117364, 'reg__num_leaves': 20, 'reg__subsample': 0.742471553380787, 'reg__colsample_bytree': 0.5922760721833944, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.530148444022808, 'reg__reg_lambda': 36.78238024179093, 'reg__min_split_gain': 2.919394390121347, 'reg__path_smooth': 1.7094085673790067, 'reg__min_child_weight': 0.0031328502113430324, 'reg__feature_fraction': 0.8310884909573368, 'reg__bagging_fraction': 0.6347902449236721, 'reg__bagging_freq': 3, 'reg__max_bin': 97, 'reg__min_data_per_group': 92}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.05198478368235849, 'reg__num_leaves': 29, 'reg__subsample': 0.7748498290482144, 'reg__colsample_bytree': 0.536819909697638, 'reg__min_child_samples': 14, 'reg__reg_alpha': 2.5874167096738403, 'reg__reg_lambda': 41.4994818548486, 'reg__min_split_gain': 3.371805942486903, 'reg__path_smooth': 0.5611831742687794, 'reg__min_child_weight': 1.6771775301568952, 'reg__feature_fraction': 0.5141298442174254, 'reg__bagging_fraction': 0.589694829088896, 'reg__bagging_freq': 3, 'reg__max_bin': 145, 'reg__min_data_per_group': 73}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.053767469151632497, 'reg__num_leaves': 17, 'reg__subsample': 0.5127971986759561, 'reg__colsample_bytree': 0.5097460749326552, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.5880125273575647, 'reg__reg_lambda': 41.99558394612644, 'reg__min_split_gain': 3.2441052204064373, 'reg__path_smooth': 0.012749258434841093, 'reg__min_child_weight': 0.0010547282380566885, 'reg__feature_fraction': 0.5139104277000701, 'reg__bagging_fraction': 0.5079856899816857, 'reg__bagging_freq': 3, 'reg__max_bin': 111, 'reg__min_data_per_group': 53}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.049668432784200856, 'reg__num_leaves': 28, 'reg__subsample': 0.8136098768058817, 'reg__colsample_bytree': 0.5995871150695843, 'reg__min_child_samples': 15, 'reg__reg_alpha': 2.882310525063665, 'reg__reg_lambda': 41.02098654979283, 'reg__min_split_gain': 3.160934541055491, 'reg__path_smooth': 0.544758280491569, 'reg__min_child_weight': 0.0011493990970659219, 'reg__feature_fraction': 0.5036136405702348, 'reg__bagging_fraction': 0.5743516500225698, 'reg__bagging_freq': 5, 'reg__max_bin': 96, 'reg__min_data_per_group': 75}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.09118782729991182, 'reg__num_leaves': 38, 'reg__subsample': 0.7402454993253493, 'reg__colsample_bytree': 0.6928031523578484, 'reg__min_child_samples': 19, 'reg__reg_alpha': 1.9854320677627988, 'reg__reg_lambda': 45.97033877068673, 'reg__min_split_gain': 3.0669071333251576, 'reg__path_smooth': 1.4457385515026888, 'reg__min_child_weight':

0.0031477003069387225, 'reg__feature_fraction': 0.5998970934595015, 'reg__bagging_fraction': 0.5381354173553824, 'reg__bagging_freq': 6, 'reg__max_bin': 169, 'reg__min_data_per_group': 89}

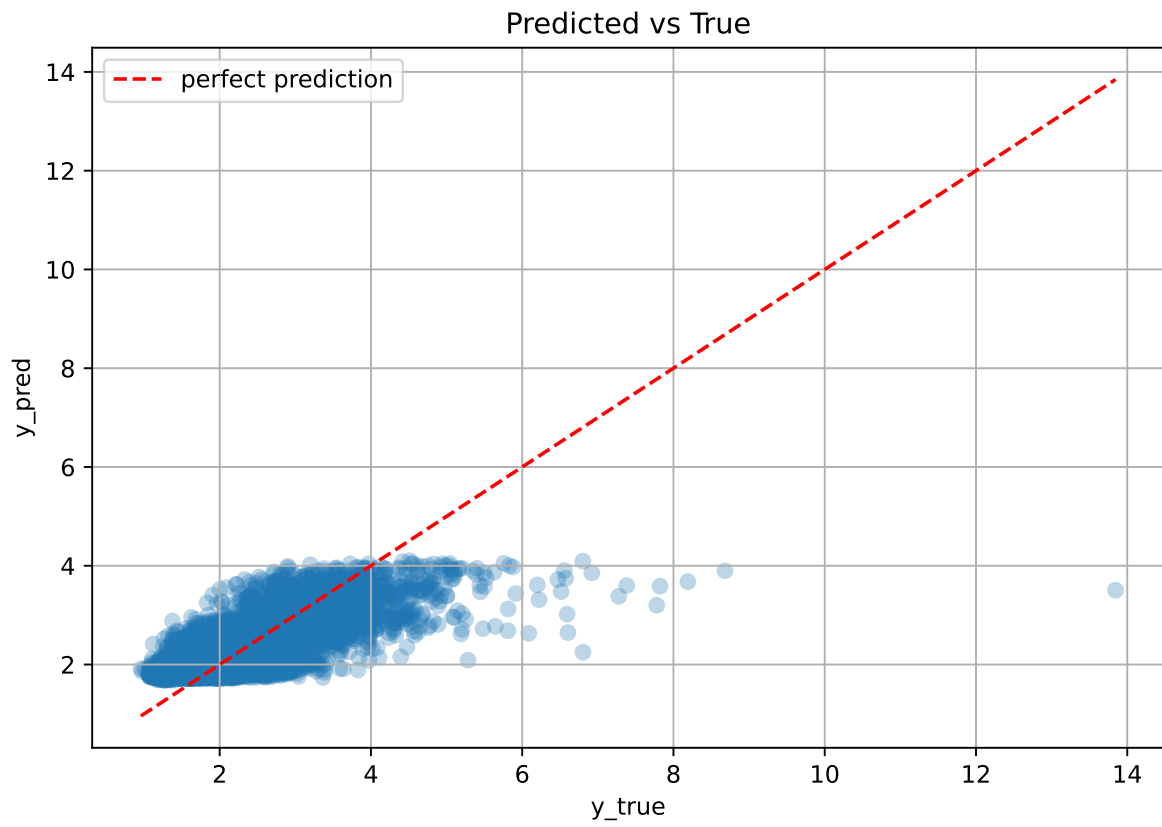
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.08782468882631791, 'reg__num_leaves': 26, 'reg__subsample': 0.8705588614050088, 'reg__colsample_bytree': 0.7314647976522666, 'reg__min_child_samples': 17, 'reg__reg_alpha': 5.055998178781506, 'reg__reg_lambda': 31.17913853700321, 'reg__min_split_gain': 3.088264010644823, 'reg__path_smooth': 1.6810482776108047, 'reg__min_child_weight': 0.005686136731292328, 'reg__feature_fraction': 0.5958124677461487, 'reg__bagging_fraction': 0.5359602127852607, 'reg__bagging_freq': 2, 'reg__max_bin': 131, 'reg__min_data_per_group': 83}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.07503752566916458, 'reg__num_leaves': 27, 'reg__subsample': 0.8383978667637514, 'reg__colsample_bytree': 0.7060225114245087, 'reg__min_child_samples': 24, 'reg__reg_alpha': 4.228374348213427, 'reg__reg_lambda': 16.675563504292178, 'reg__min_split_gain': 4.41639774084512, 'reg__path_smooth': 1.2113295902594914, 'reg__min_child_weight': 3.858050868361951, 'reg__feature_fraction': 0.5416469046235315, 'reg__bagging_fraction': 0.5592260330308992, 'reg__bagging_freq': 5, 'reg__max_bin': 226, 'reg__min_data_per_group': 67}

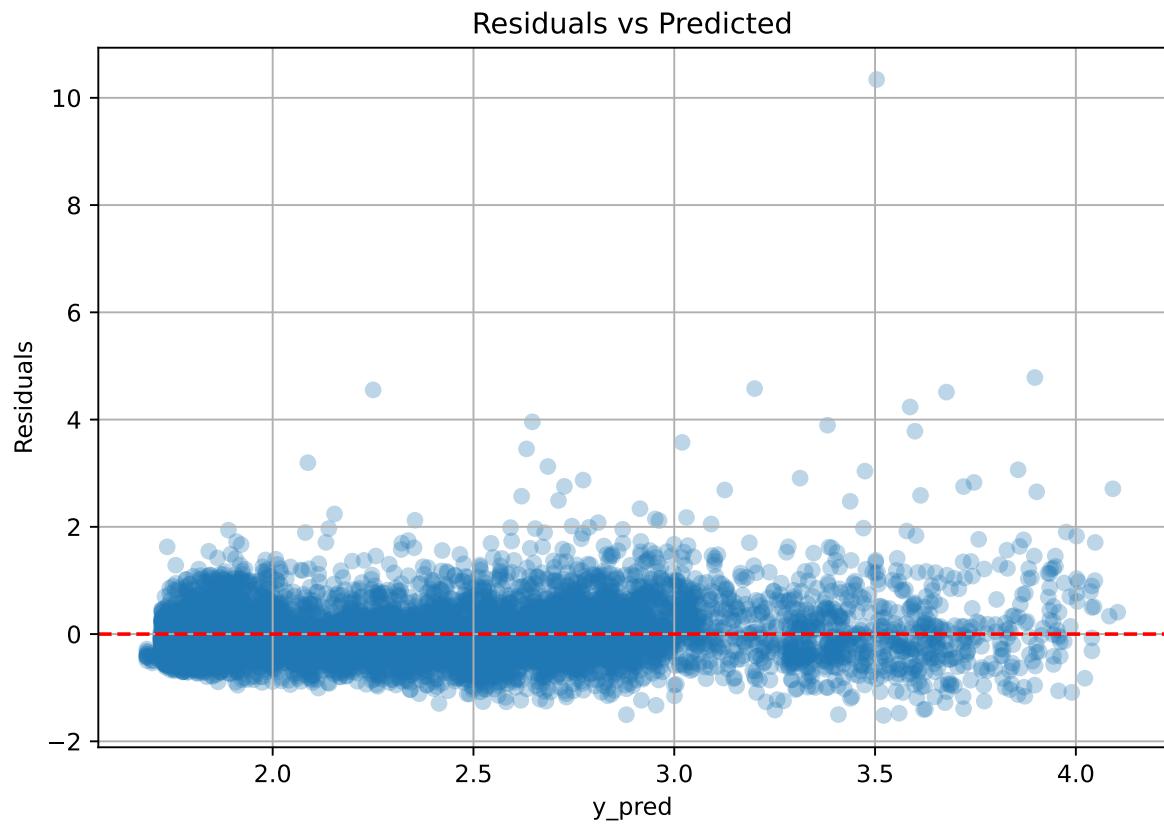
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.05859192698268176, 'reg__num_leaves': 16, 'reg__subsample': 0.665097278017093, 'reg__colsample_bytree': 0.516802877762669, 'reg__min_child_samples': 17, 'reg__reg_alpha': 7.747079643186126, 'reg__reg_lambda': 35.85801957134318, 'reg__min_split_gain': 1.9752933606675822, 'reg__path_smooth': 0.840704774802617, 'reg__min_child_weight': 0.4881757068961316, 'reg__feature_fraction': 0.7469326045504117, 'reg__bagging_fraction': 0.5296048551140762, 'reg__bagging_freq': 6, 'reg__max_bin': 209, 'reg__min_data_per_group': 64}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.059544349873505736, 'reg__num_leaves': 15, 'reg__subsample': 0.7815161258280217, 'reg__colsample_bytree': 0.5892223687883708, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.875550438467739, 'reg__reg_lambda': 17.460614741327753, 'reg__min_split_gain': 4.878602208664326, 'reg__path_smooth': 0.07496803511481565, 'reg__min_child_weight': 7.908765842976292, 'reg__feature_fraction': 0.5082610122095399, 'reg__bagging_fraction': 0.6014637294797283, 'reg__bagging_freq': 7, 'reg__max_bin': 210, 'reg__min_data_per_group': 50}

3.5.2 Scatter



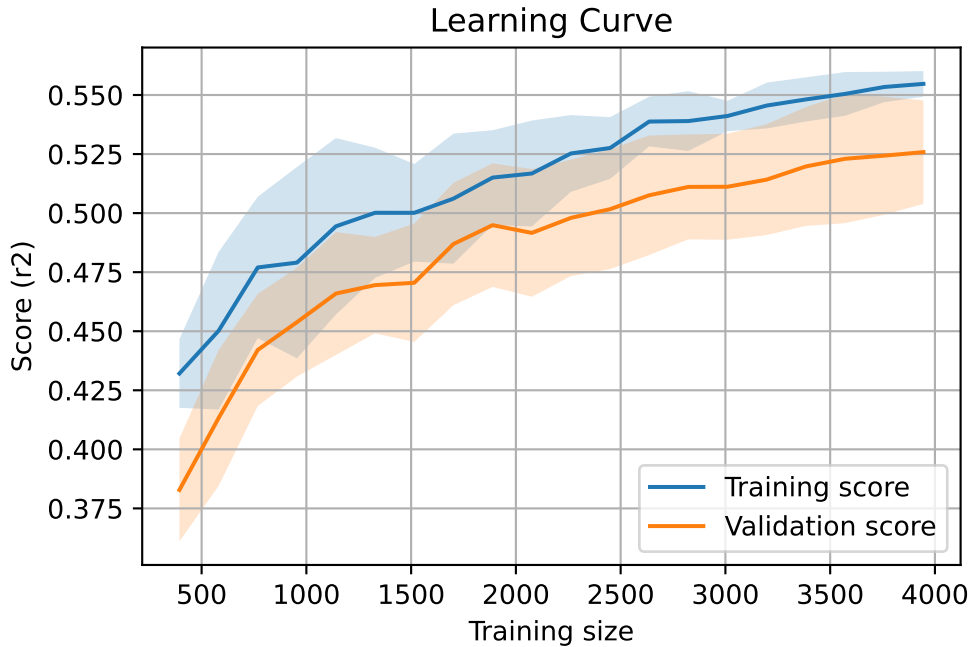
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8052, `Len(training)` : 5637,

`Len(training_real)` : 3946, `Len(test_of_training)` : 1691, `Len(test)` : 2415

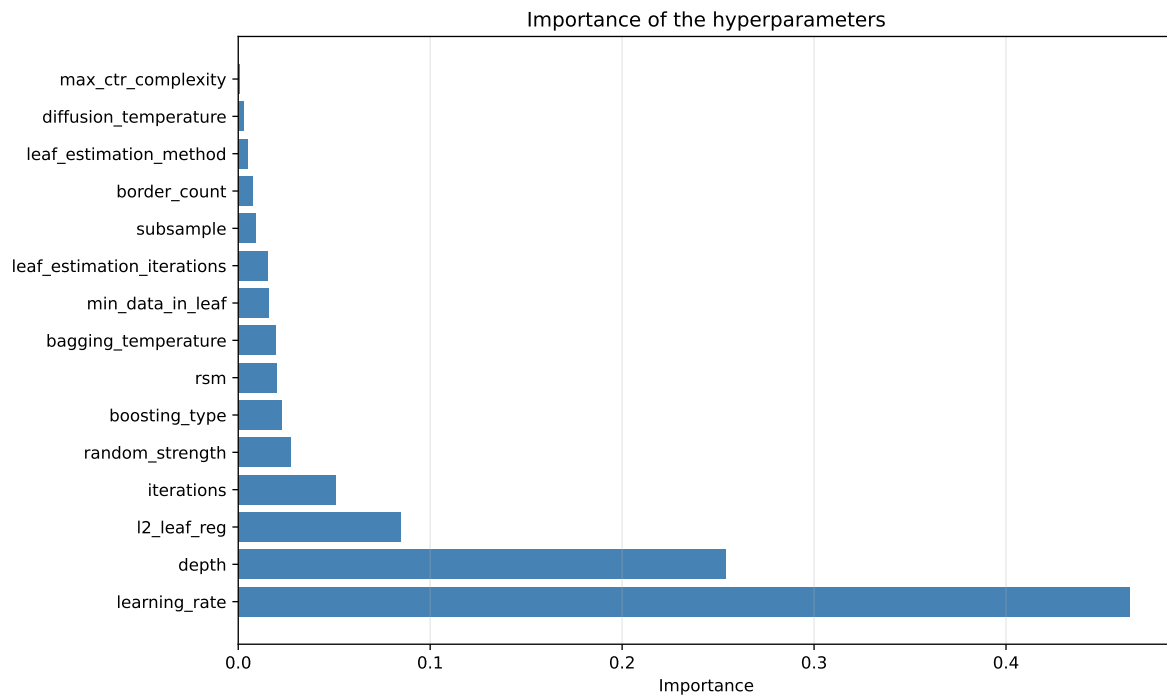
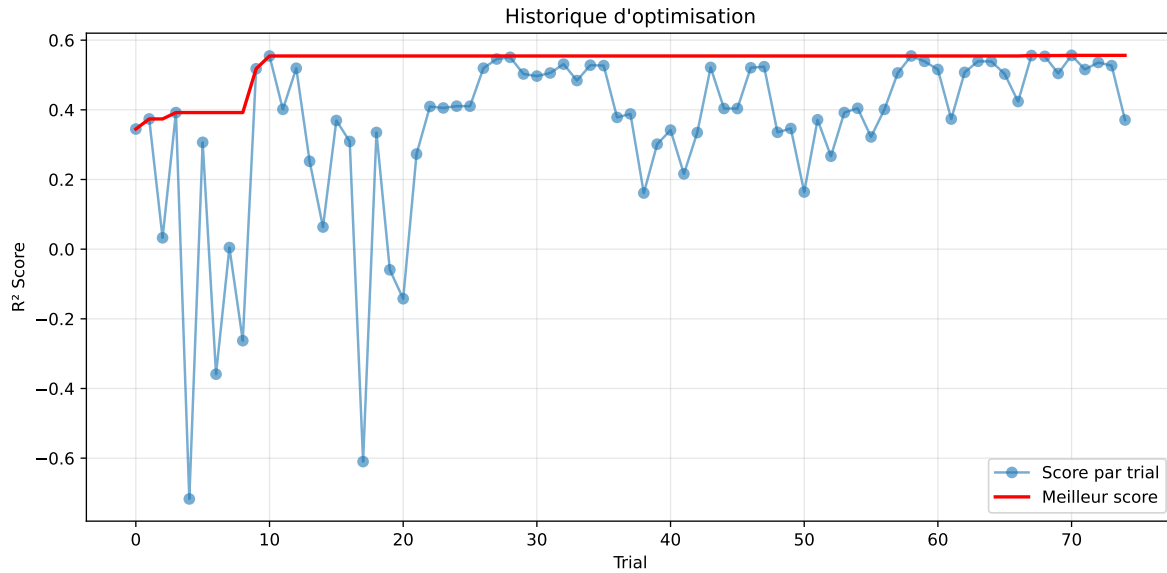


3.6 Model : CatBoost

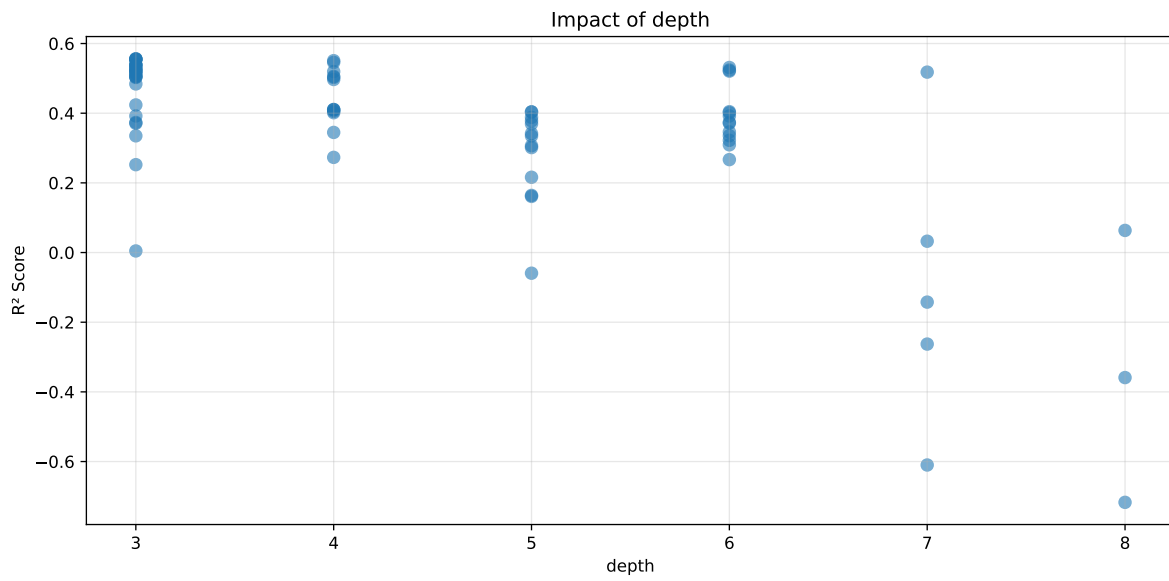
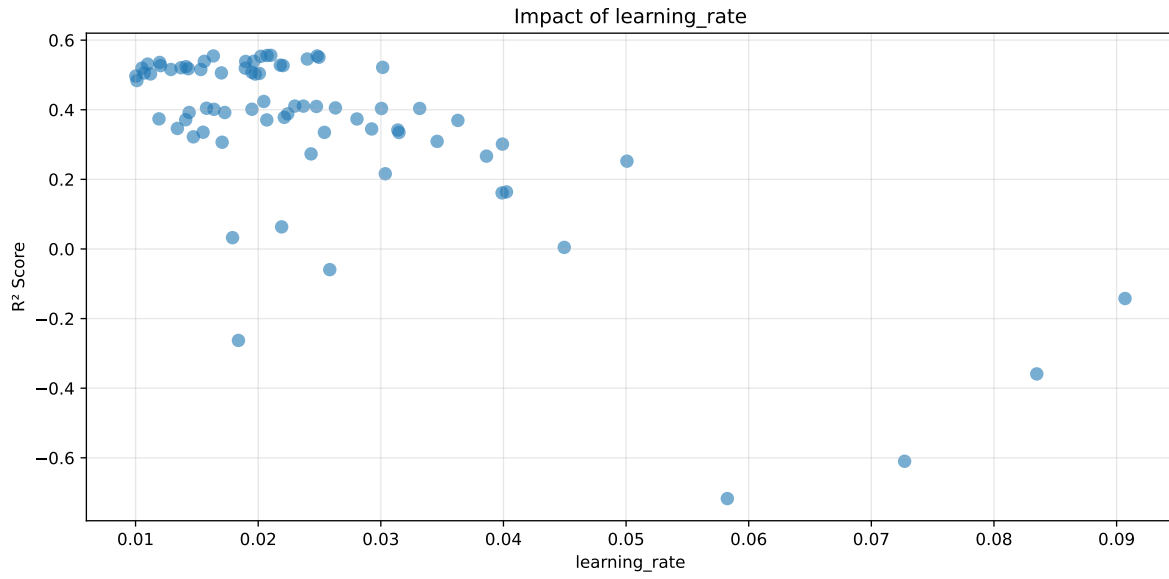
3.6.1 Gridsearch

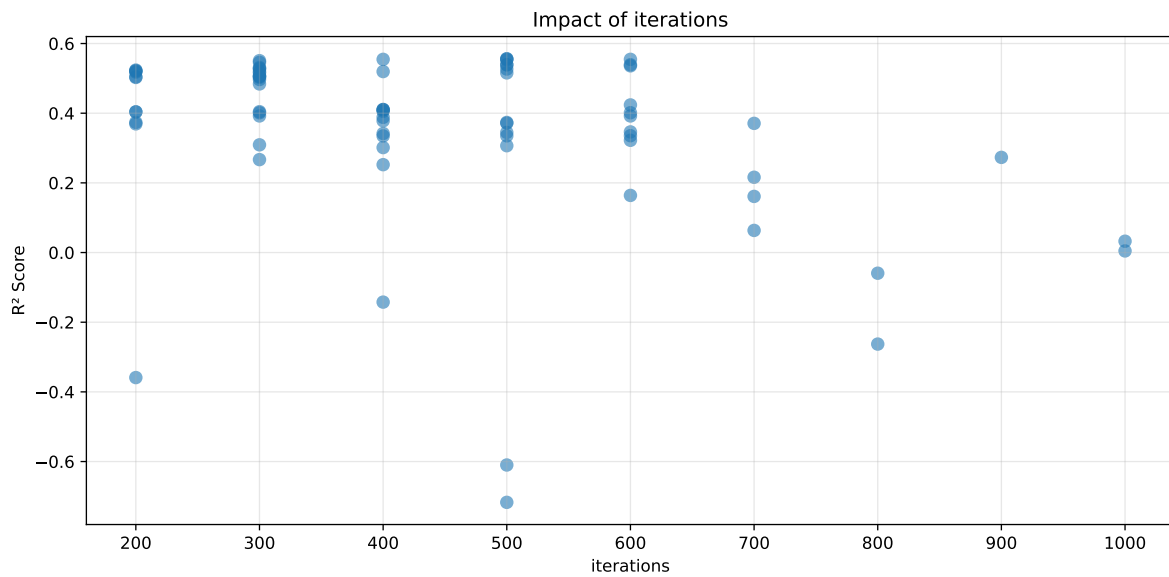
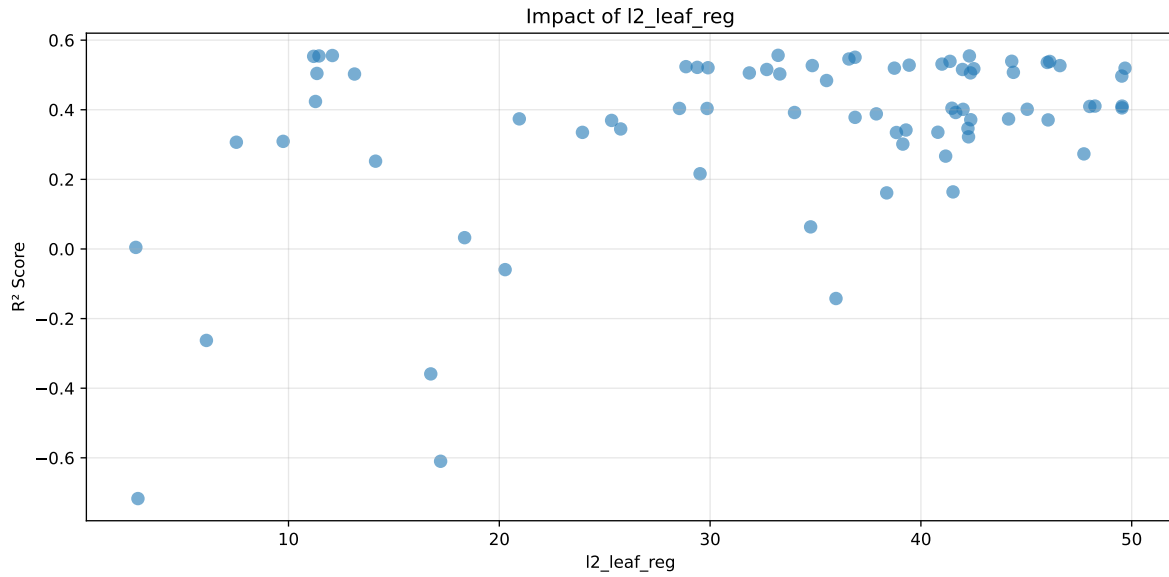
Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.804, min=0.959, max=13.845
 y_{test} : mean=2.369, std=0.782, min=0.973, max=7.822

===== TOP Importance FEATURES =====
 feature importance $r_umidity$ 34.852441 $r_temperature_fft_3h_fft_power1$ 21.810097 r_NH3 9.975893 r_CO2 9.483799 $r_temperature$ 8.074344 $r_umidity_fft_3h_fft_power1$ 4.612193 $r_NH3_fft_3h_fft_power1$ 3.470401 $r_temperature_fft_3h_fft_f1_cph$ 3.377158 r_THI 2.921532 $r_umidity_fft_3h_fft_f1_cph$ 0.856158 $r_NH3_fft_3h_fft_f1_cph$ 0.565983



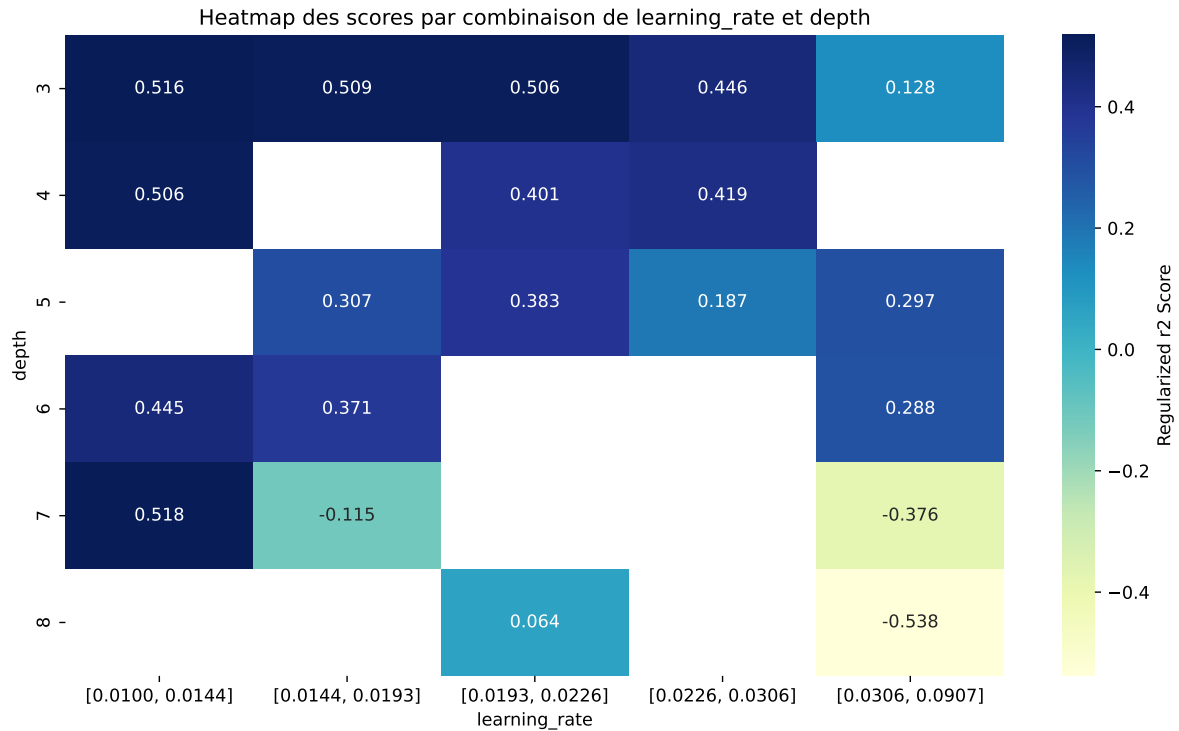
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'l2_leaf_reg', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5824	0.5564	0.5591	0.0225	0.5564	0.5634	0.007	1.0126	1.0469
2	0.5832	0.5558	0.563	0.0203	0.5558	0.5637	0.0079	1.0141	1.0493
3	0.5813	0.5546	0.5604	0.0195	0.5546	0.5646	0.01	1.018	1.0481
4	0.5762	0.5546	0.5593	0.0211	0.5546	0.5642	0.0096	1.0173	1.039
5	0.575	0.5535	0.5561	0.0205	0.5535	0.5595	0.006	1.0109	1.0388
6	0.5719	0.5508	0.553	0.0219	0.5508	0.5565	0.0056	1.0102	1.0383
7	0.5695	0.5458	0.5494	0.0207	0.5458	0.5543	0.0084	1.0155	1.0434
8	0.557	0.5393	0.5414	0.0197	0.5393	0.5454	0.0061	1.0113	1.0329
9	0.5563	0.5391	0.5407	0.0212	0.5391	0.5448	0.0057	1.0105	1.0319
10	0.5562	0.5388	0.5421	0.0211	0.5388	0.5459	0.0071	1.0131	1.0323
---	---	---	---	---	---	---	---	---	---
75	0.9558	0.677	0.6863	0.0217	-0.7171	0.7073	0.0303	1.0447	1.4119

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.02102768174454546, 'reg__l2_leaf_reg': 33.225459805625675, 'reg__bagging_temperature': 0.4166065734704101, 'reg__random_strength': 2.933221531854967, 'reg__subsample': 0.7460343323219168, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7711213827023536, 'reg__border_count': 35, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6202.745617463598}

Rank 2: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.020721639716101827, 'reg__l2_leaf_reg': 12.080710270983356, 'reg__bagging_temperature': 4.963140207368476, 'reg__random_strength': 2.8961612218007744, 'reg__subsample': 0.7127739101710768, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.723944327947267, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6292.776004104104}

Rank 3: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.016343630652100247, 'reg__l2_leaf_reg': 42.30035822404166, 'reg__bagging_temperature': 2.3121042302966326, 'reg__random_strength': 1.7618249235856491, 'reg__subsample': 0.6929991558695526, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8240500384730822, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8636.86440228436}

Rank 4: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.024806294578033343, 'reg__l2_leaf_reg': 11.445094269751849, 'reg__bagging_temperature': 1.4483516356174957, 'reg__random_strength': 2.437888882081257, 'reg__subsample': 0.5535139928298479, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7040355017258424, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6932.942579535538}

Rank 5: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.020214251432914214, 'reg__l2_leaf_reg': 11.193088260542968, 'reg__bagging_temperature': 4.383371538035256, 'reg__random_strength': 2.888378008562978, 'reg__subsample': 0.7109600139999639, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7197773710924088, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6185.742885947799}

Rank 6: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.024954014563882125, 'reg__l2_leaf_reg': 36.87576808778795, 'reg__bagging_temperature': 1.4877936970818104, 'reg__random_strength': 2.7672358449628898, 'reg__subsample': 0.6098432476032621, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7955073069827142, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6073.333150600859}

Rank 7: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.024001193850322605, 'reg__l2_leaf_reg': 36.58491184589369, 'reg__bagging_temperature': 1.6368214278749282, 'reg__random_strength': 2.764599821856477, 'reg__subsample': 0.6079414295320015,

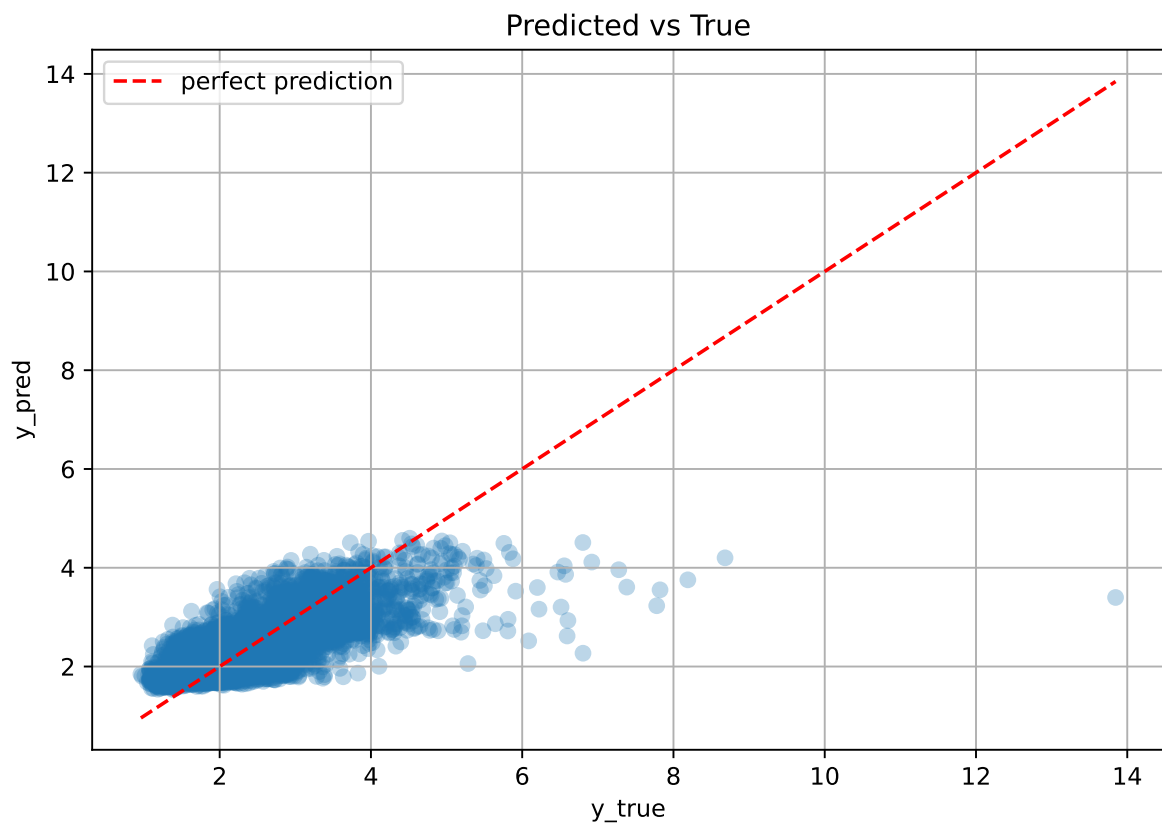
'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7843676600340227, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5571.862573255866}

Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01964816037885371, 'reg__l2_leaf_reg': 44.308565716395314, 'reg__bagging_temperature': 4.957615245189107, 'reg__random_strength': 4.479692312106562, 'reg__subsample': 0.8965363063269469, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7801638600337655, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3657.7904542349916}

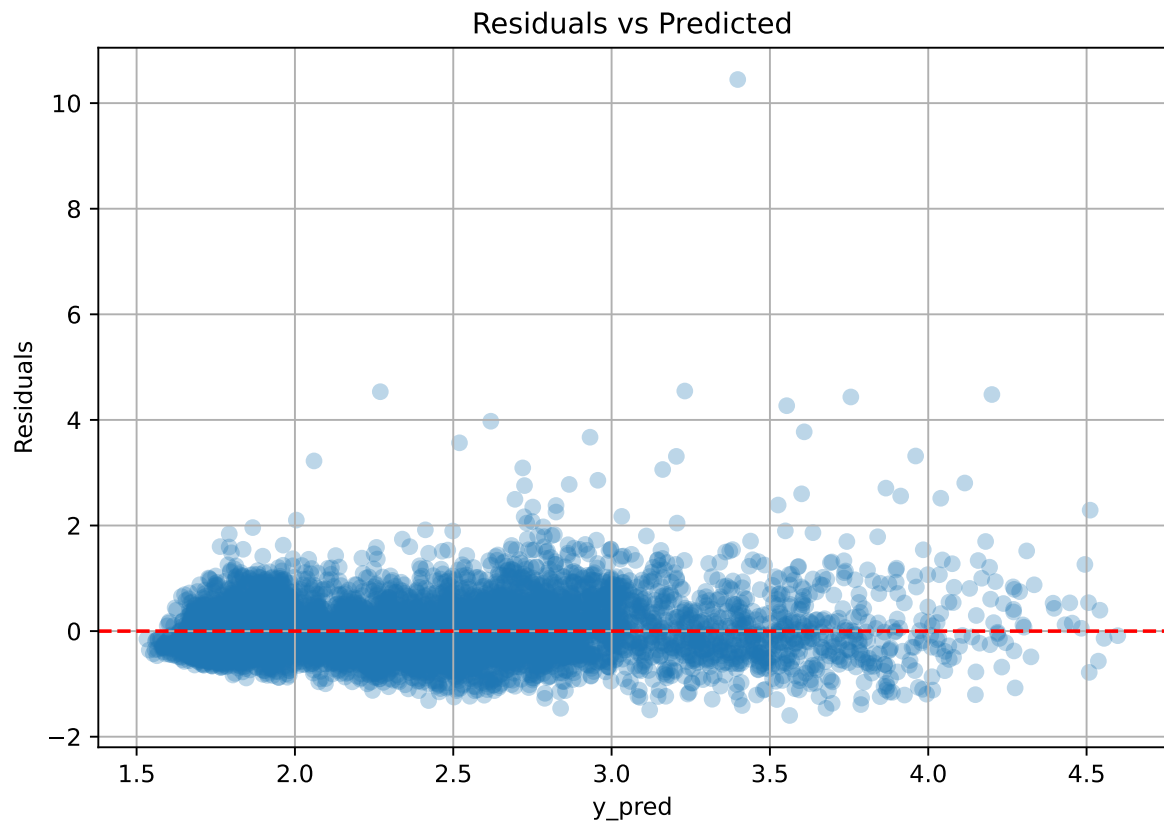
Rank 9: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.0156050950691422, 'reg__l2_leaf_reg': 41.379858962851834, 'reg__bagging_temperature': 3.8950800806399952, 'reg__random_strength': 3.666012574908757, 'reg__subsample': 0.7374654371521518, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6952897785932549, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8699.13341230206}

Rank 10: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01897904586058874, 'reg__l2_leaf_reg': 46.110990155754926, 'reg__bagging_temperature': 4.60446532349159, 'reg__random_strength': 3.6069473105584255, 'reg__subsample': 0.6290165148733747, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.778721293092922, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3797.4640418569634}

3.6.2 Scatter



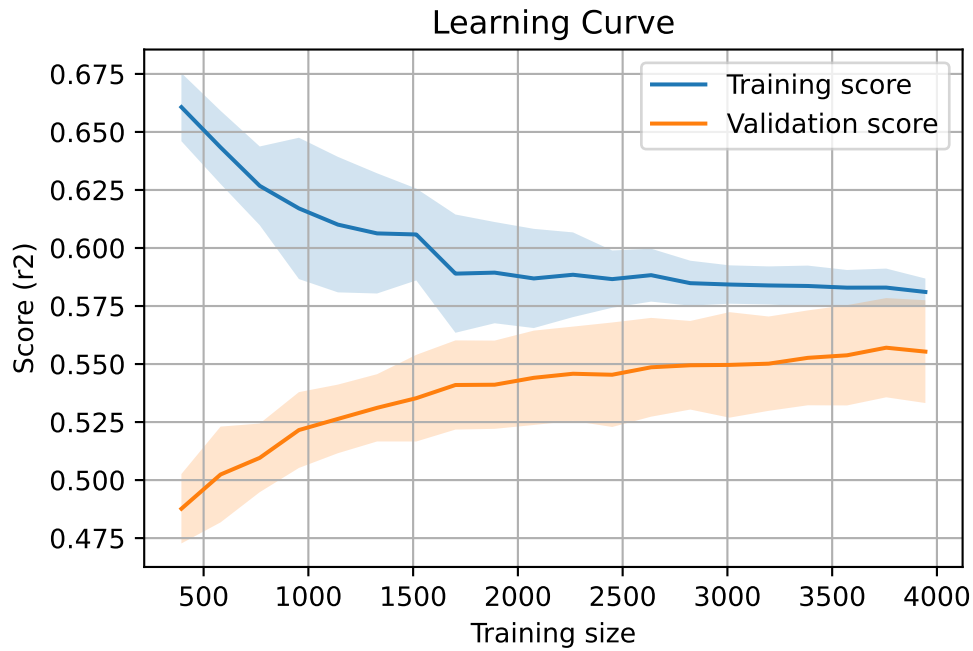
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8052, `Len(training)` : 5637,

`Len(training_real)` : 3946, `Len(test_of_training)` : 1691, `Len(test)` : 2415



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.457	0.455	0.026	0.462	0.463	0.008	1.017	1.004	nan
LGBM	0.553	0.529	0.021	0.531	0.546	0.018	1.034	1.046	0.5286
CatBoost	0.582	0.556	0.022	0.559	0.563	0.007	1.013	1.047	0.5564

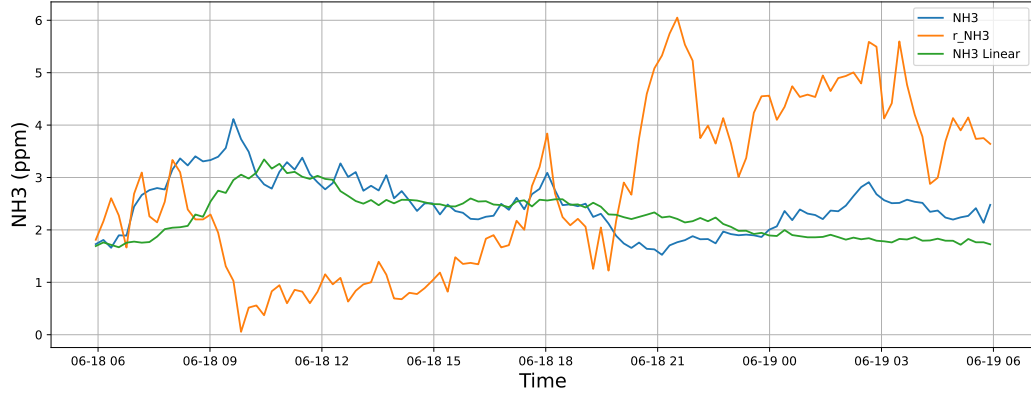
4 Plot

4.1 Standalone plots

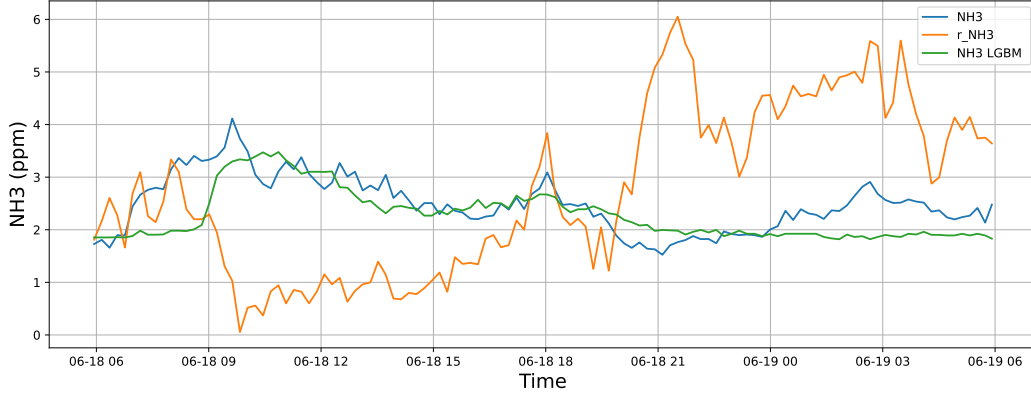
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

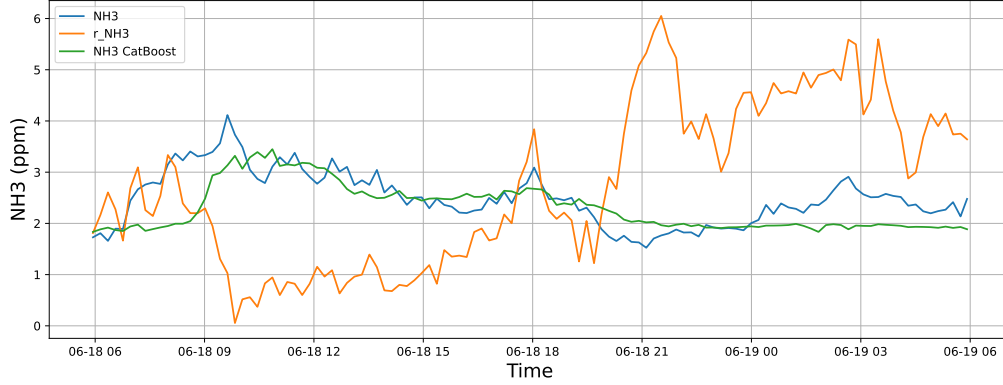
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

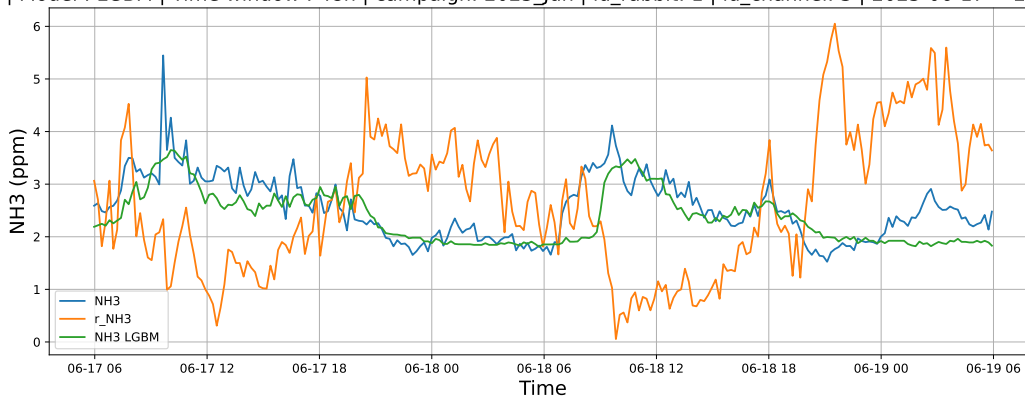


4.1.1.2 Time window : 48

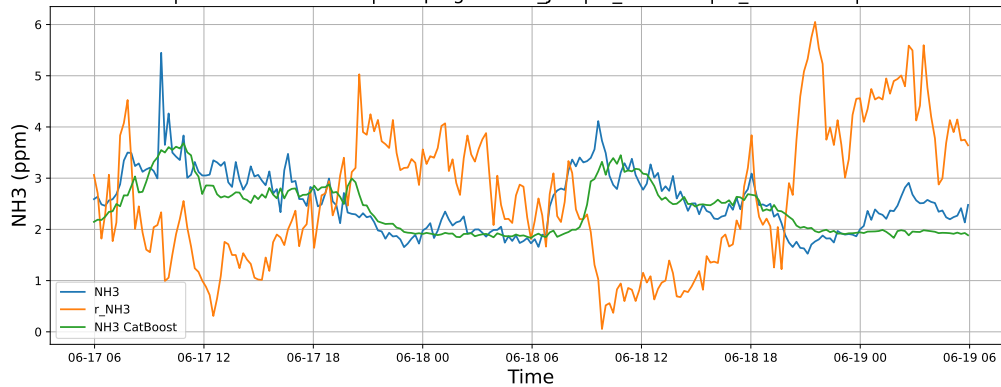
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

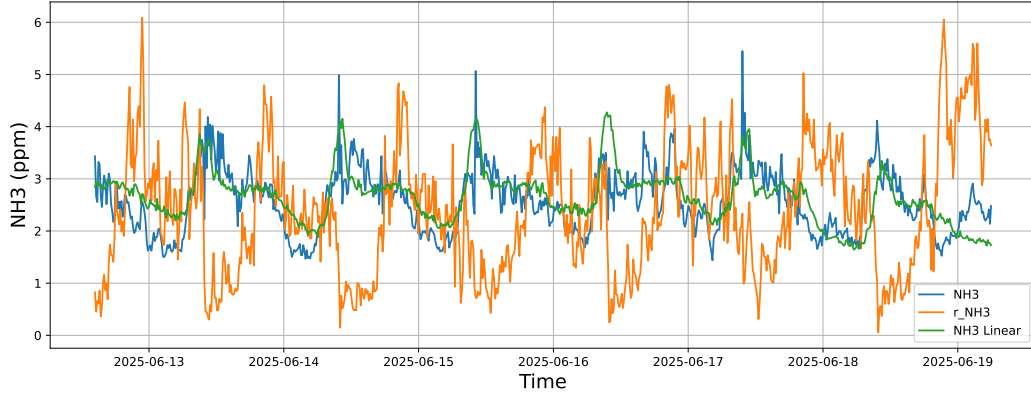


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

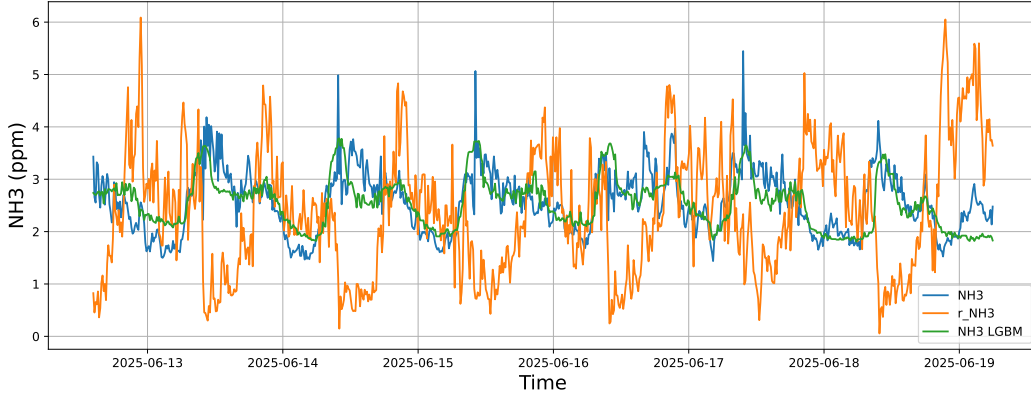


4.1.1.3 Time window : ALL

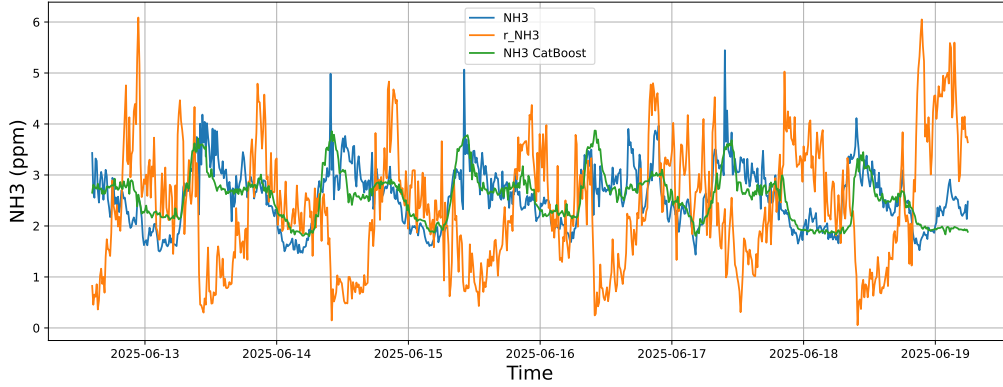
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



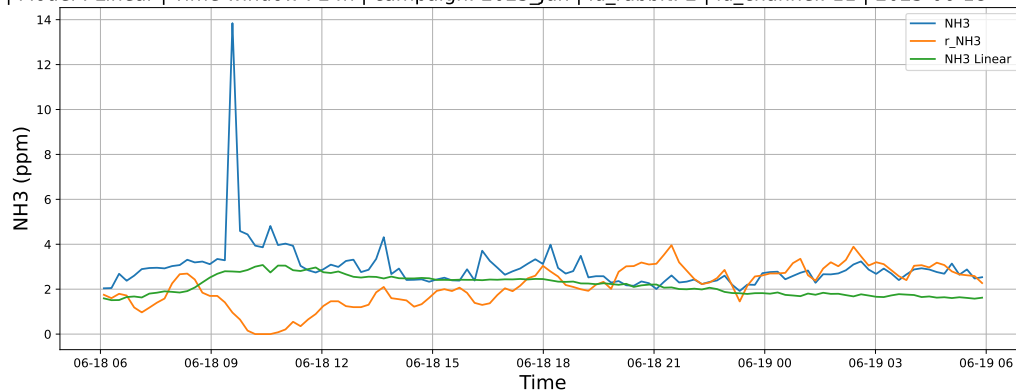
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



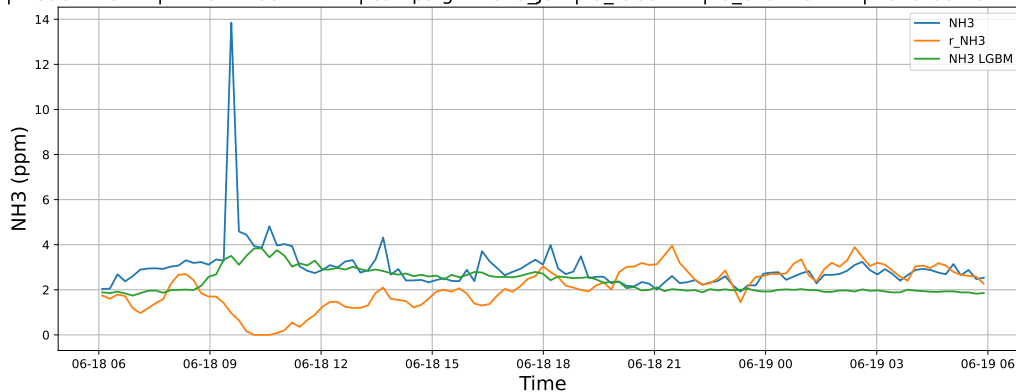
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

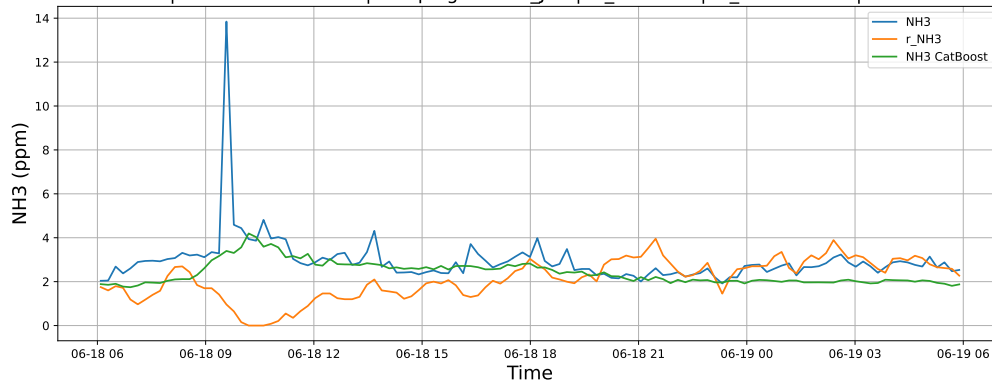
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

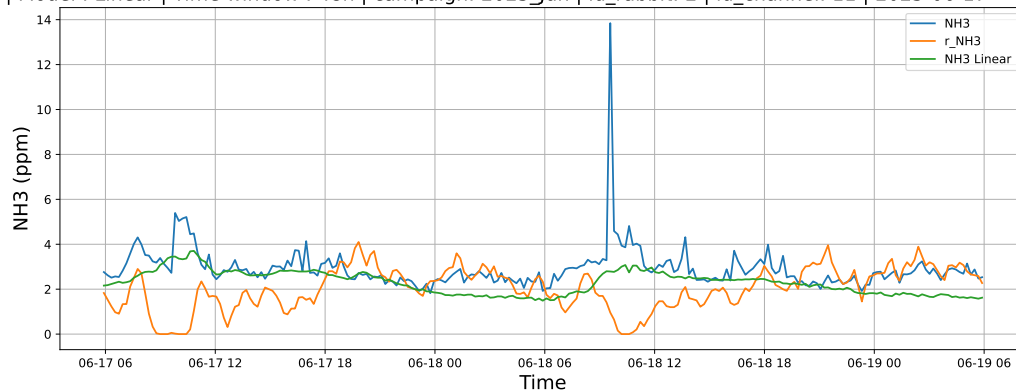


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

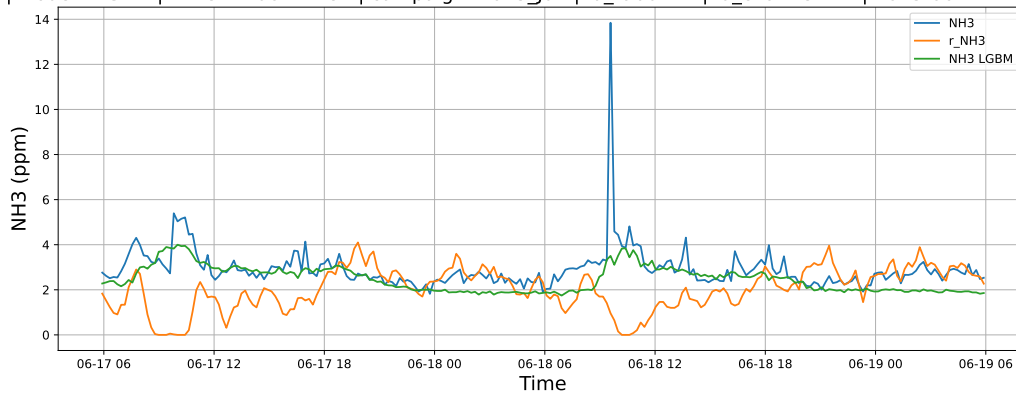


4.1.2.2 Time window : 48

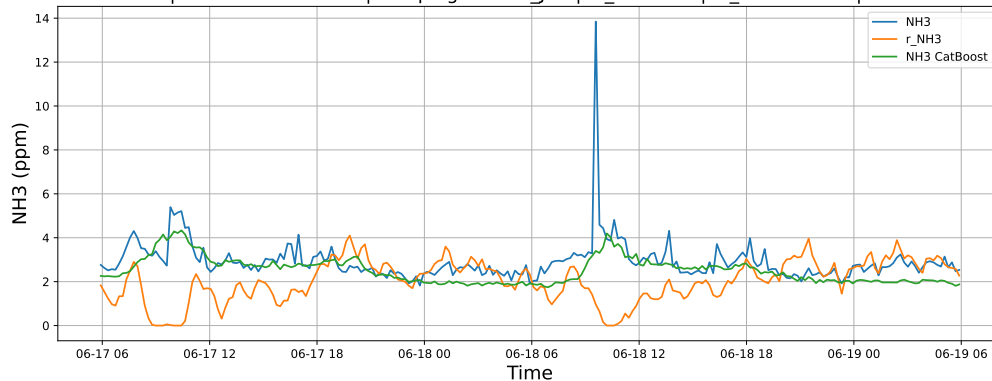
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

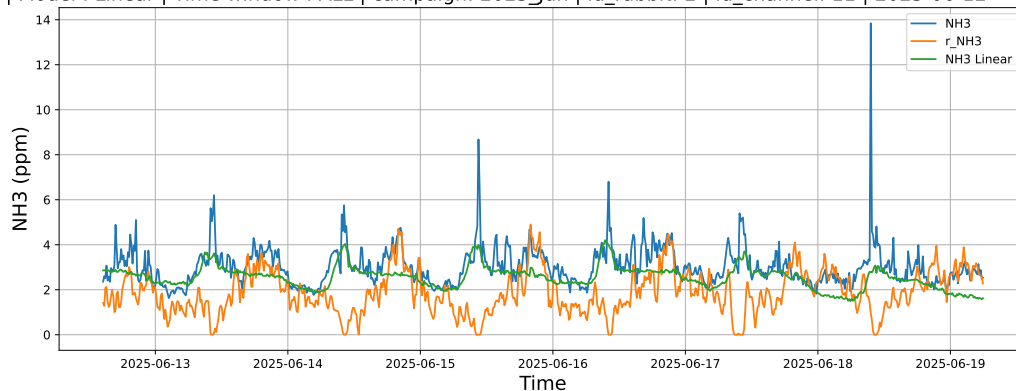


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

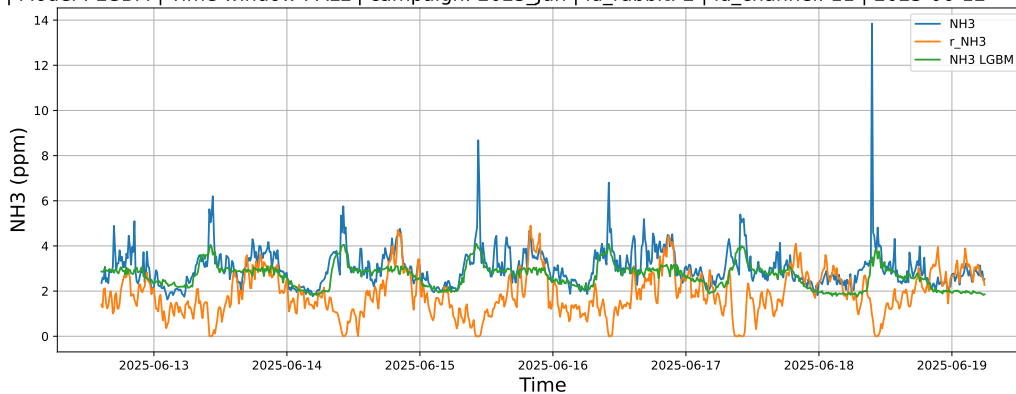


4.1.2.3 Time window : ALL

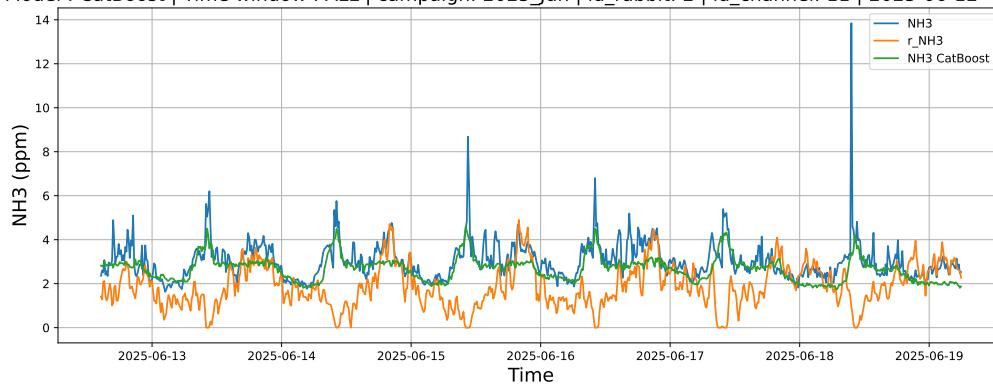
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



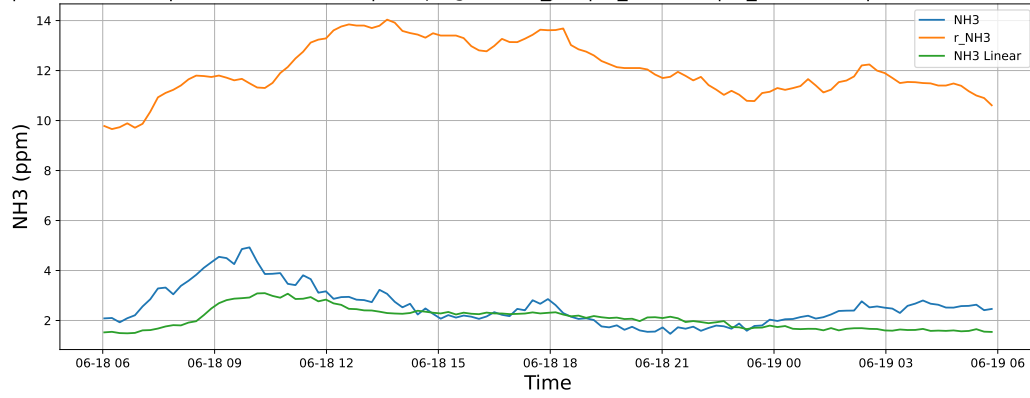
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



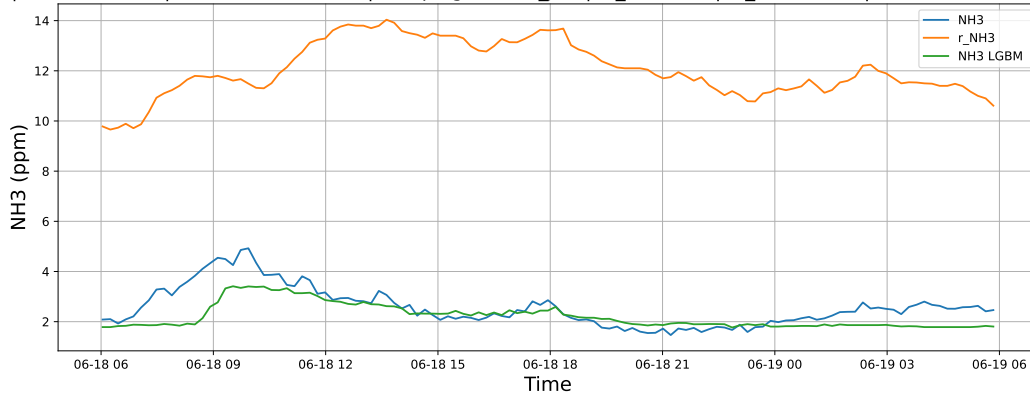
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

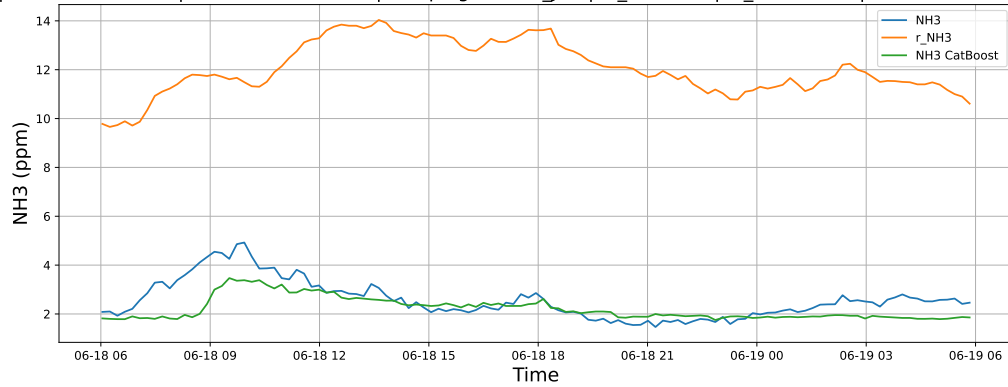
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

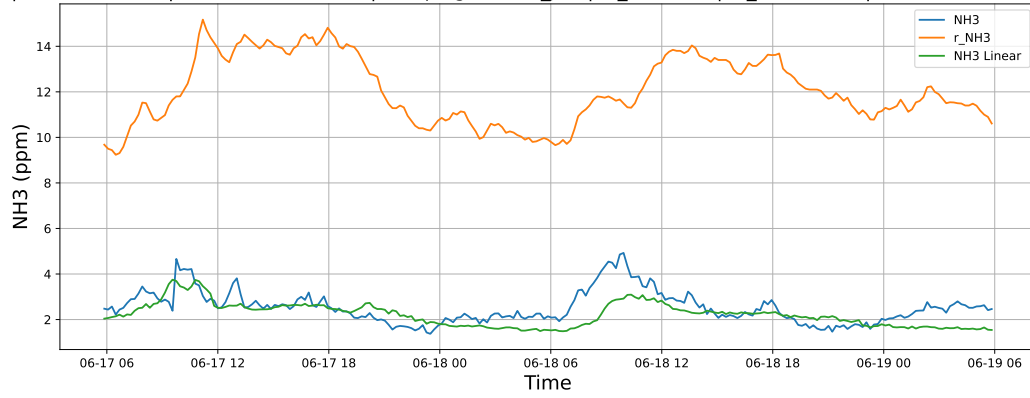


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

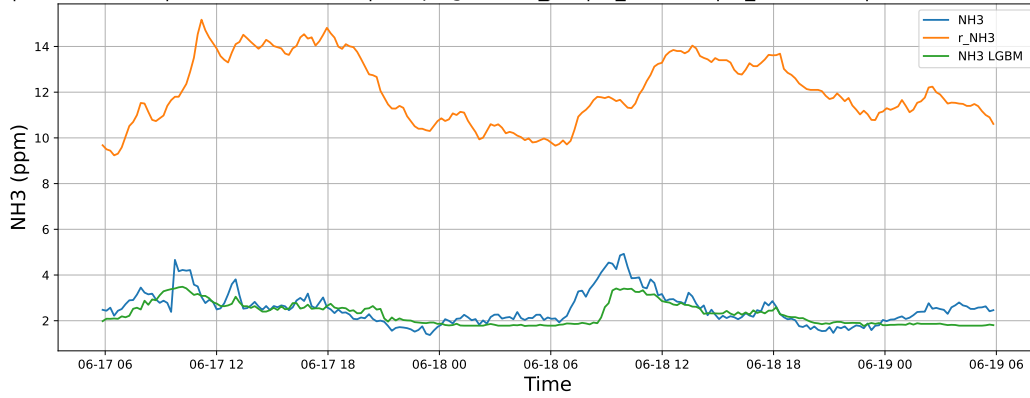


4.1.3.2 Time window : 48

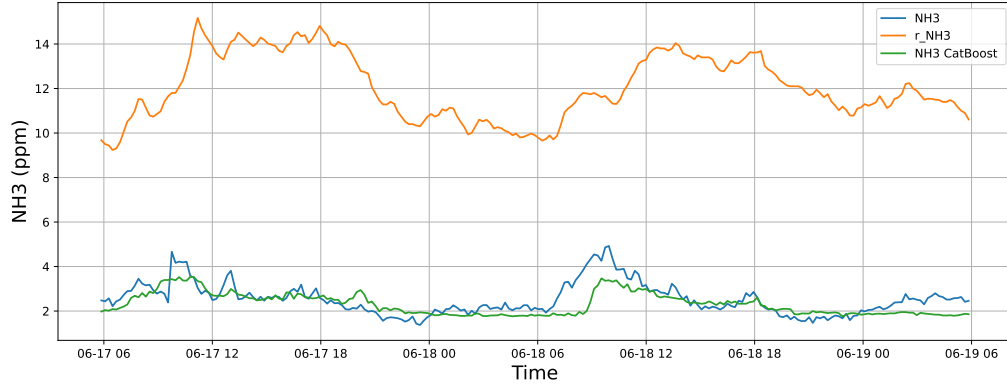
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

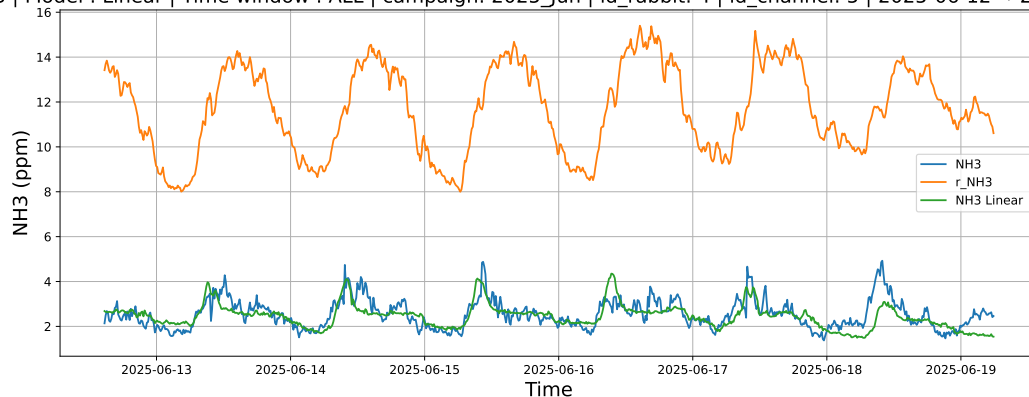


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

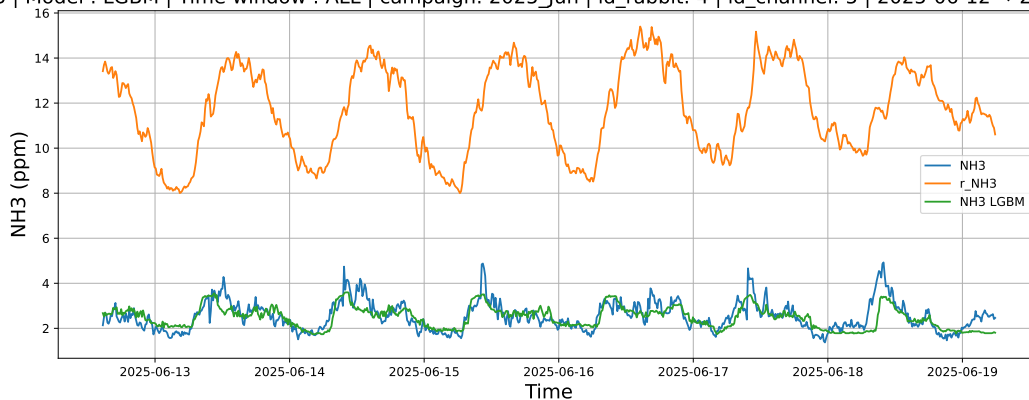


4.1.3.3 Time window : ALL

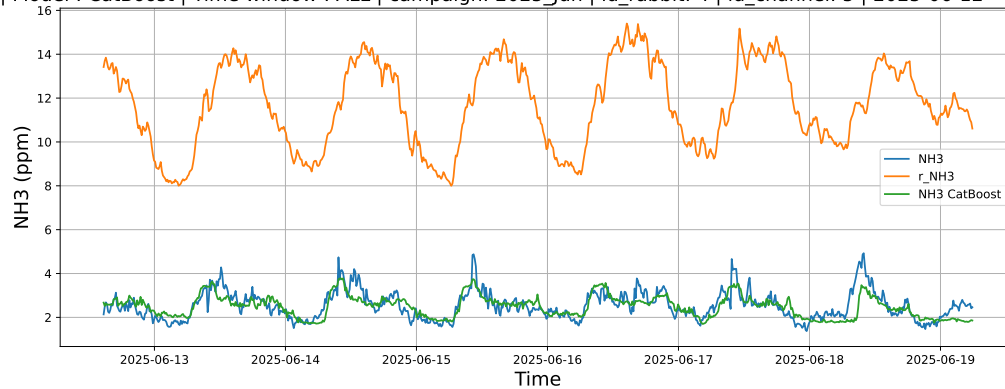
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



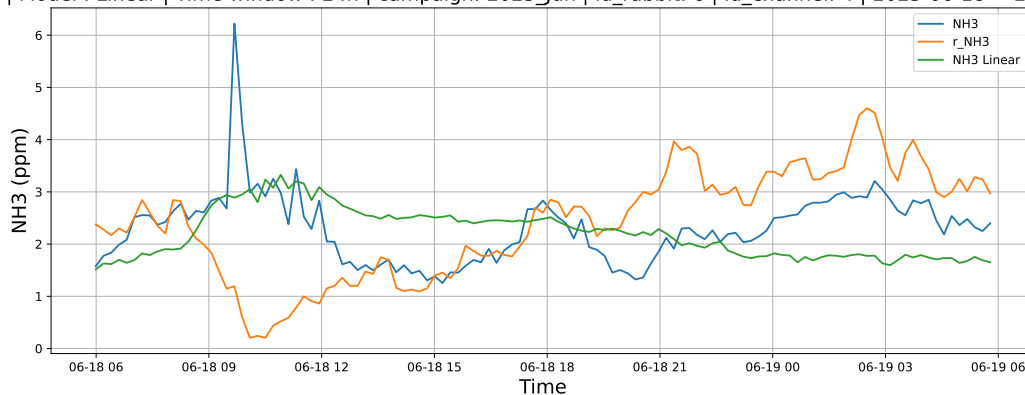
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



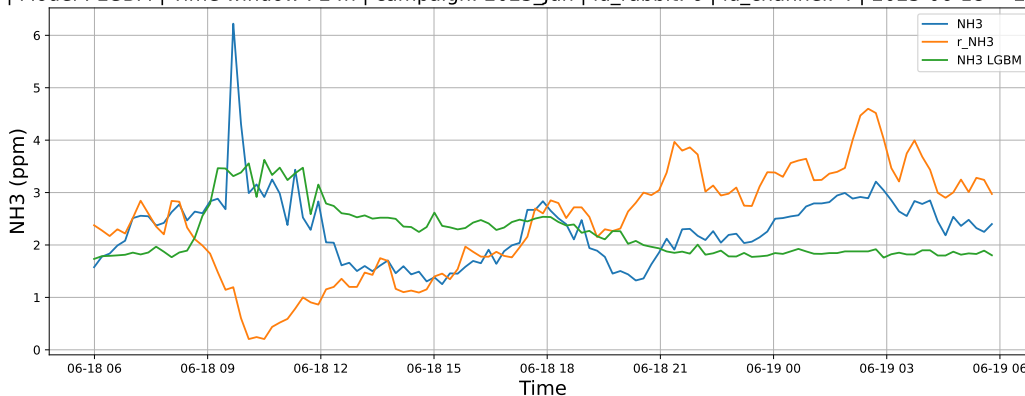
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

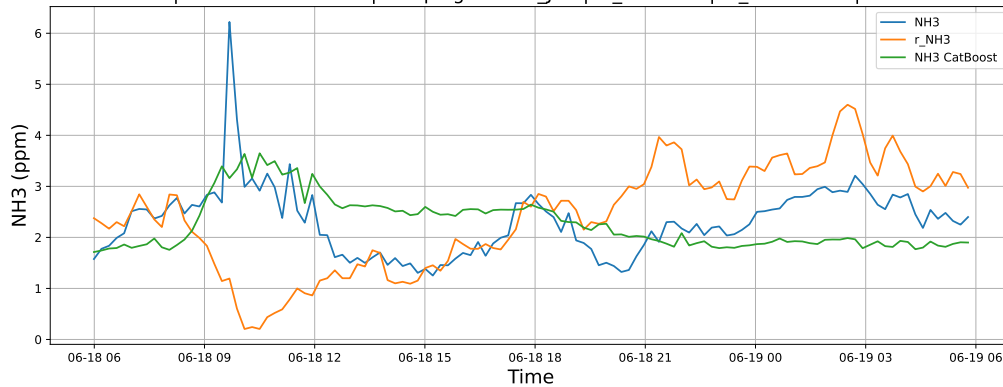
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

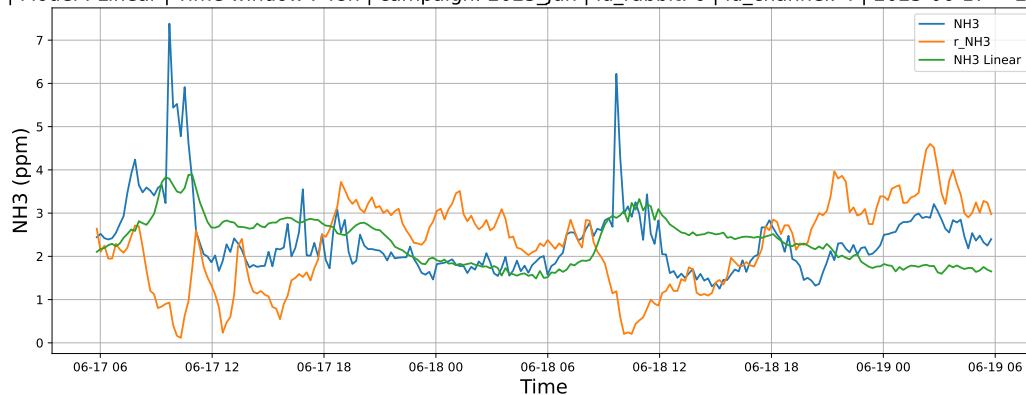


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

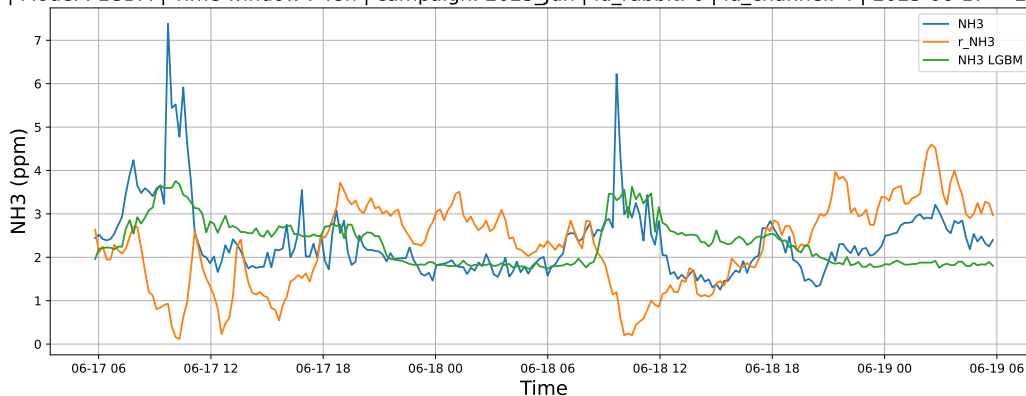


4.1.4.2 Time window : 48

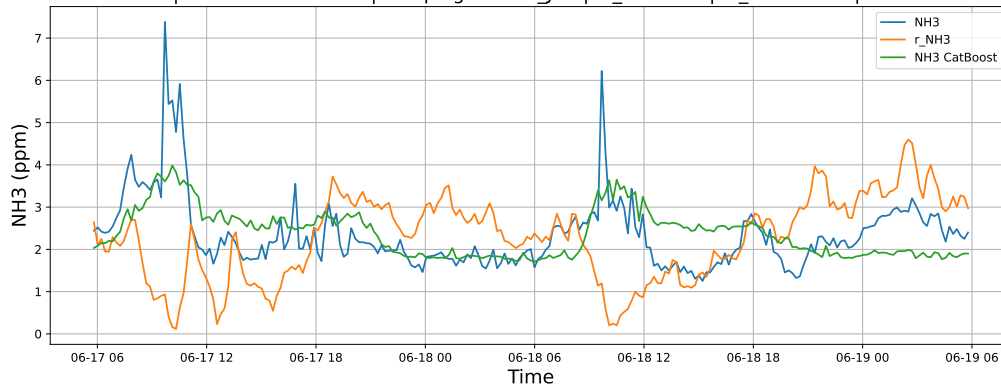
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

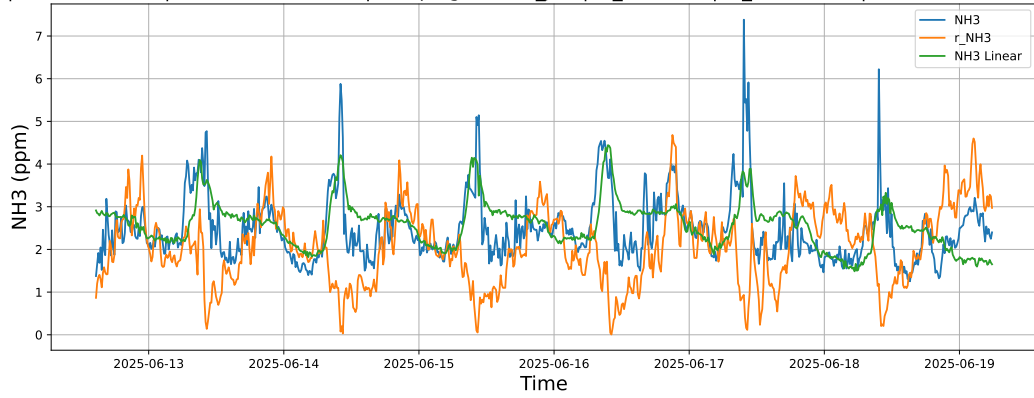


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

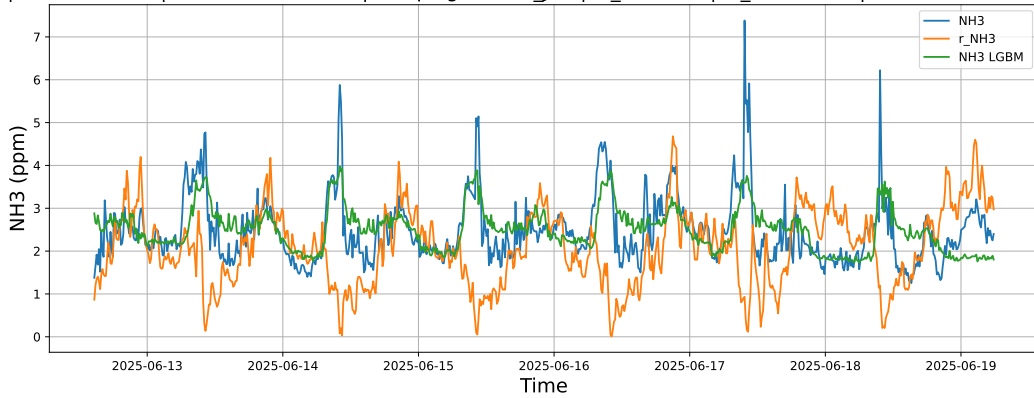


4.1.4.3 Time window : ALL

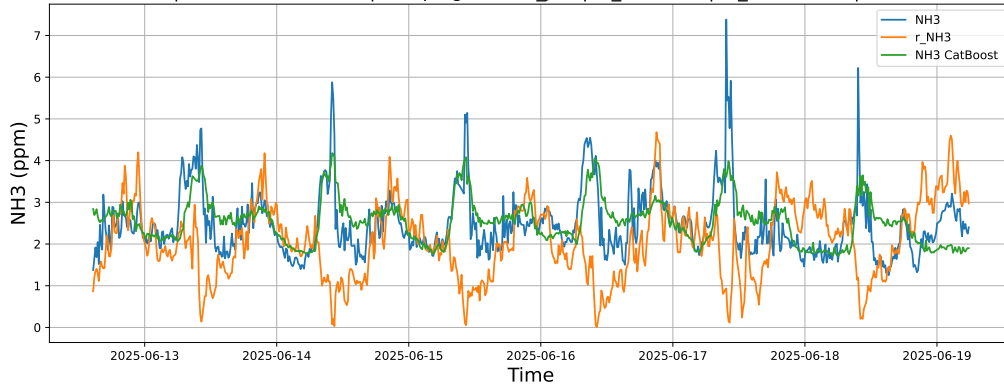
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



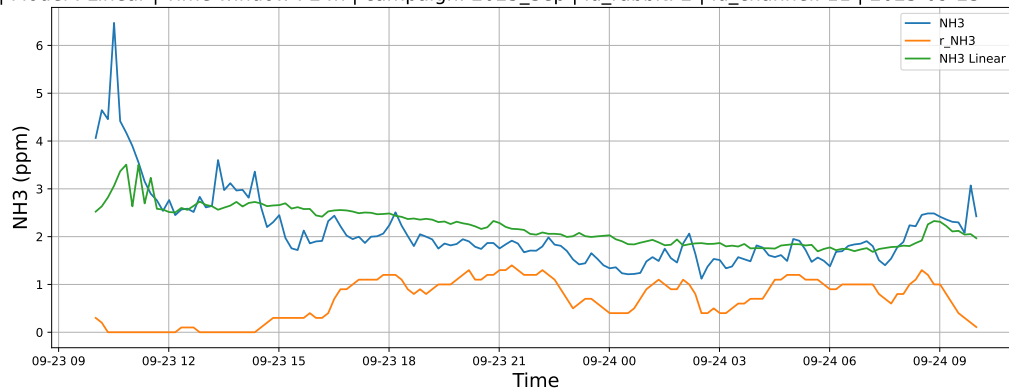
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



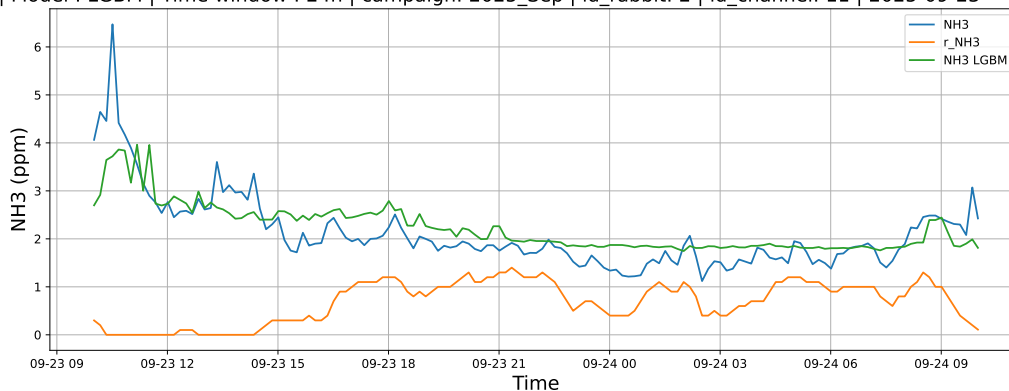
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

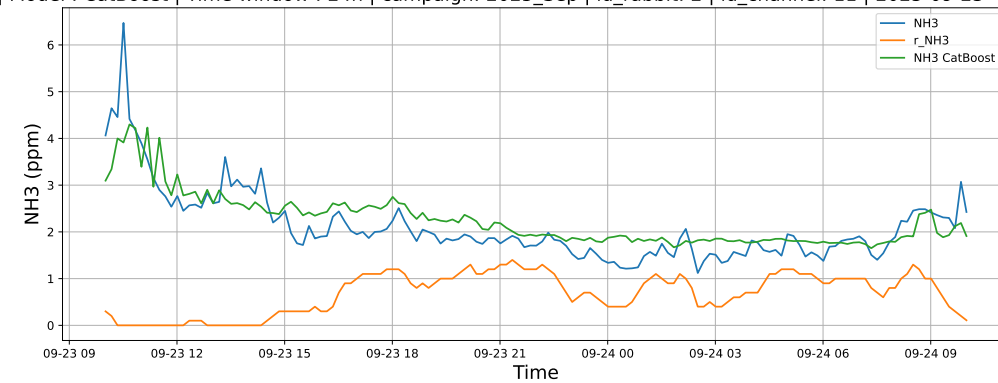
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

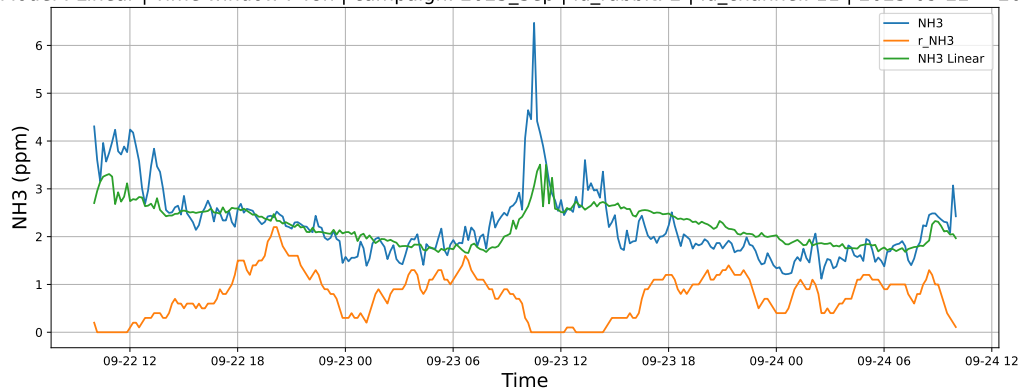


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

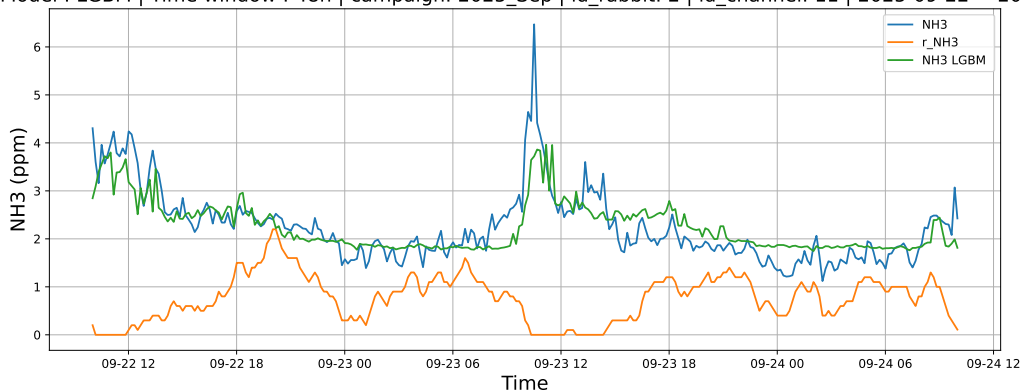


4.1.5.2 Time window : 48

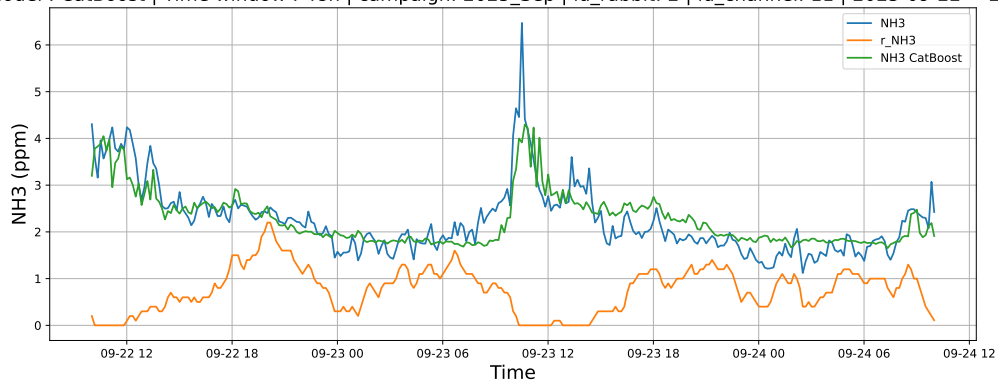
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

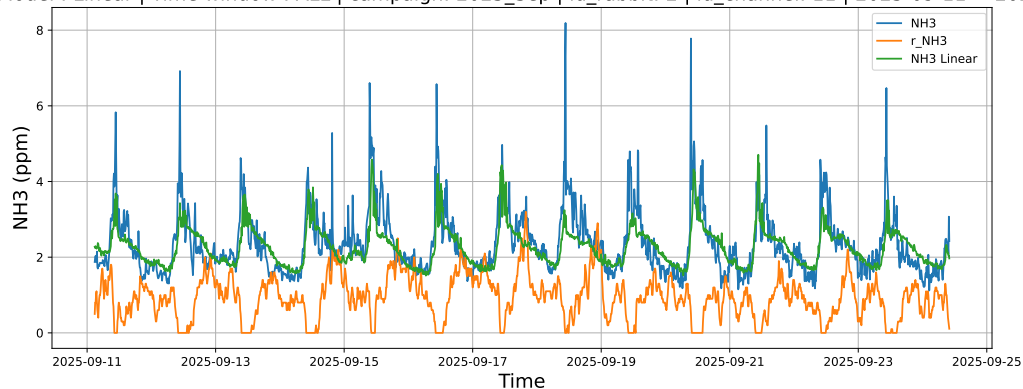


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

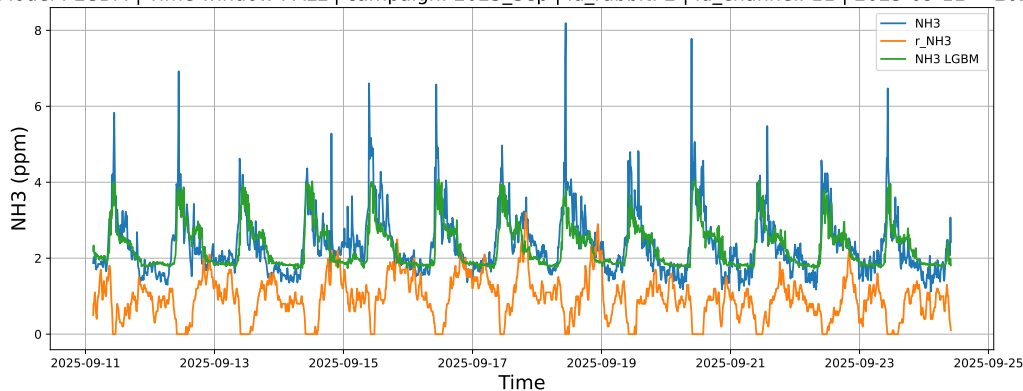


4.1.5.3 Time window : ALL

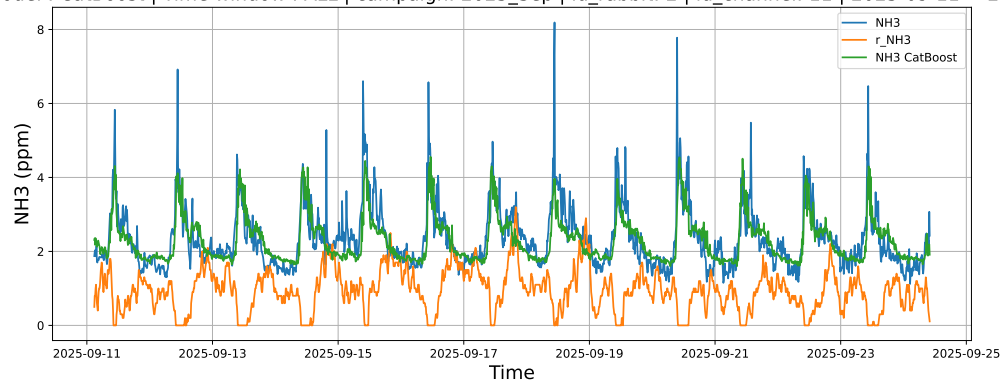
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



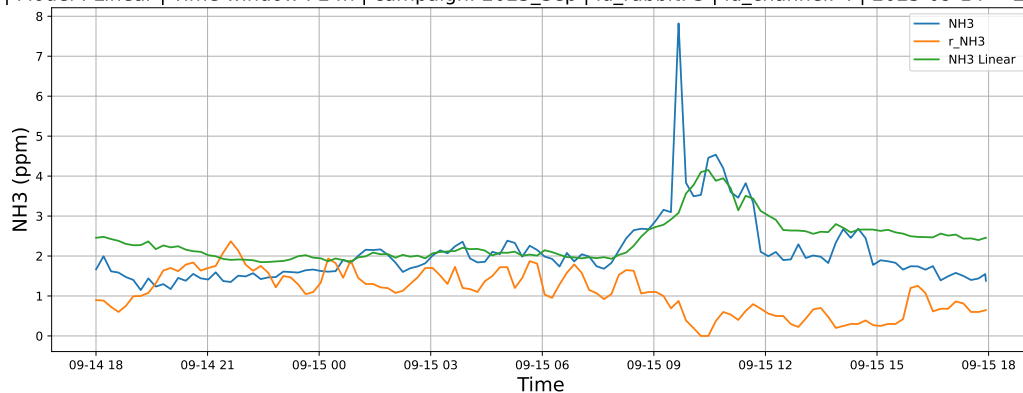
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



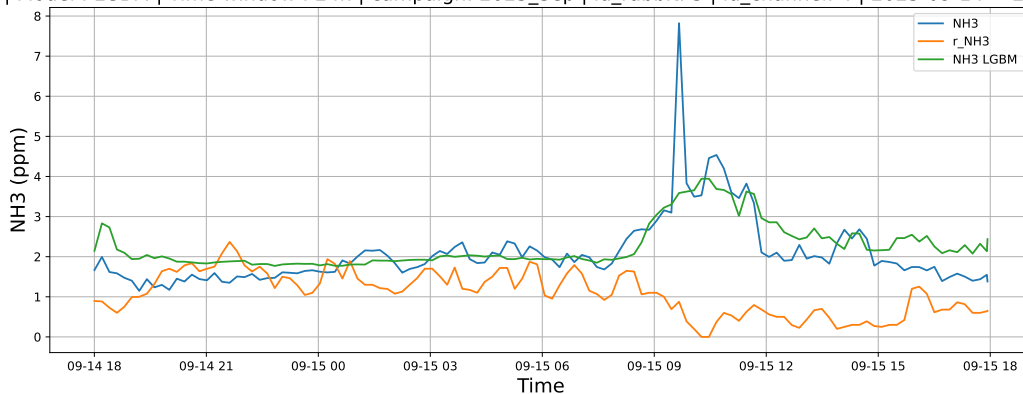
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

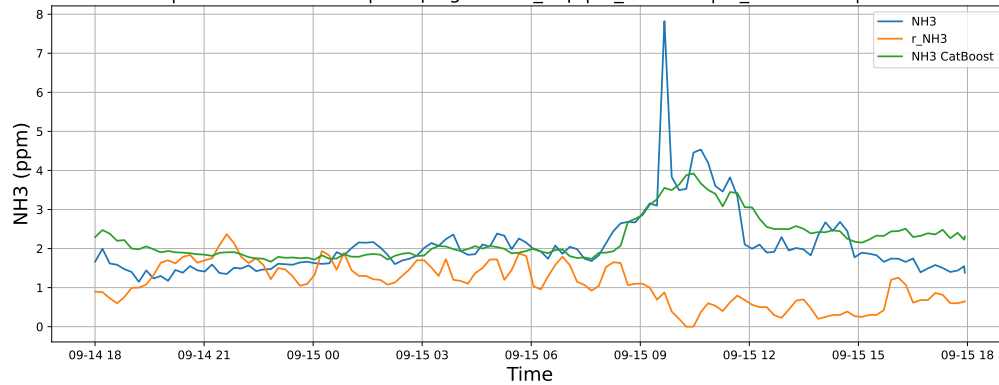
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

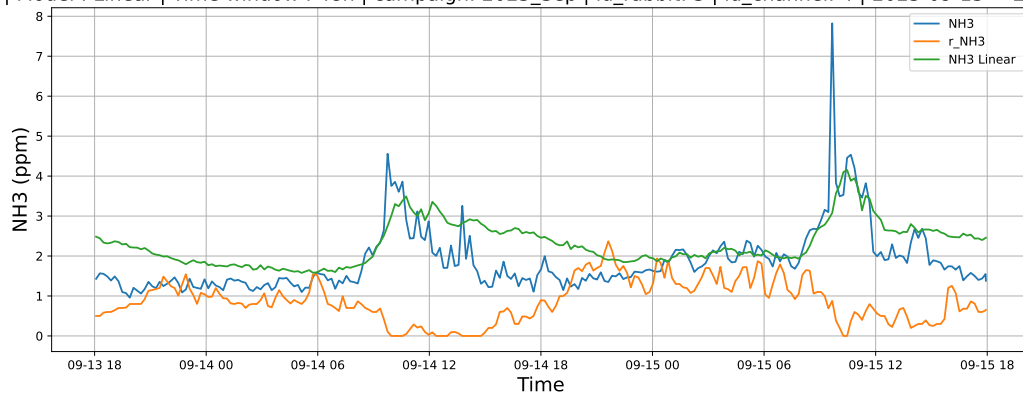


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

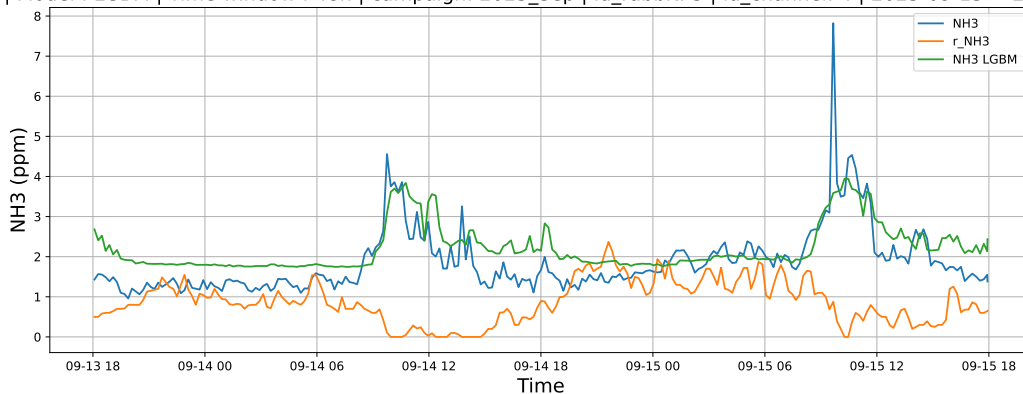


4.1.6.2 Time window : 48

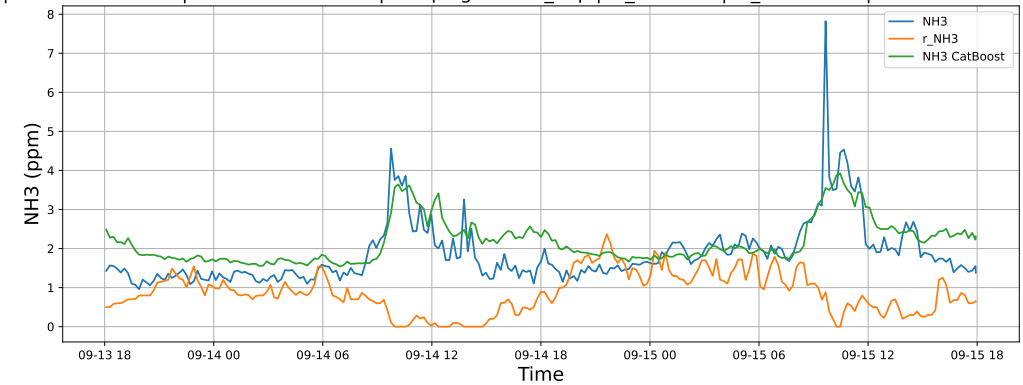
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

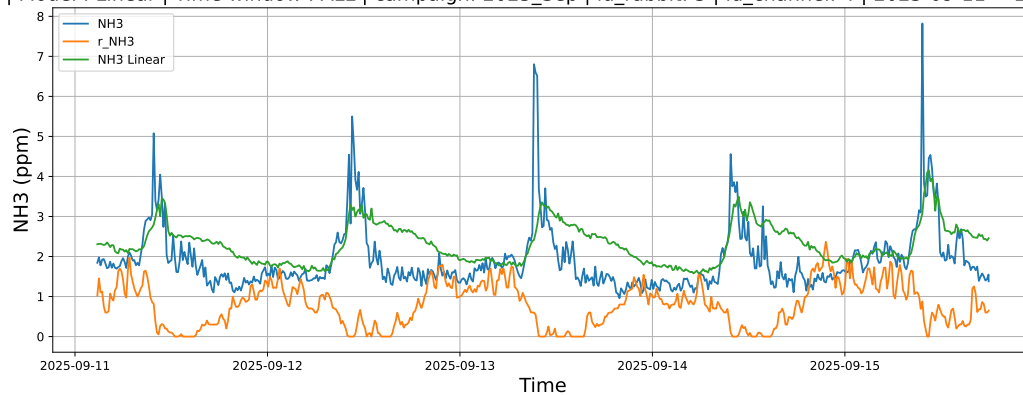


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

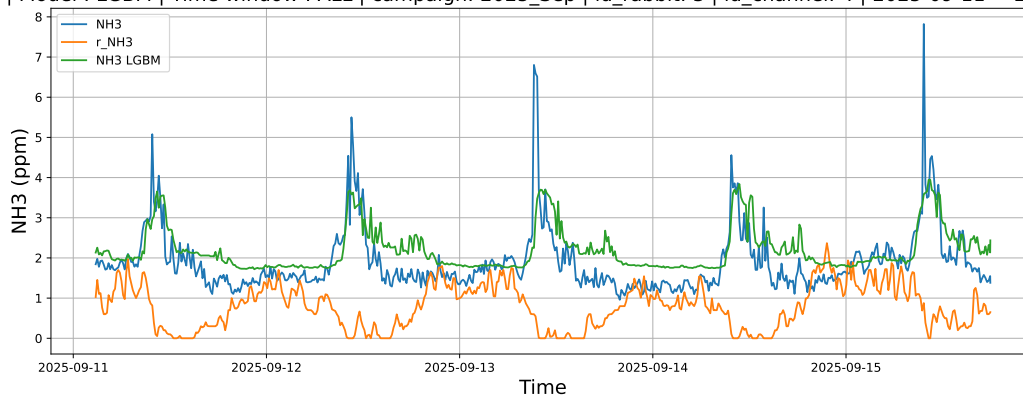


4.1.6.3 Time window : ALL

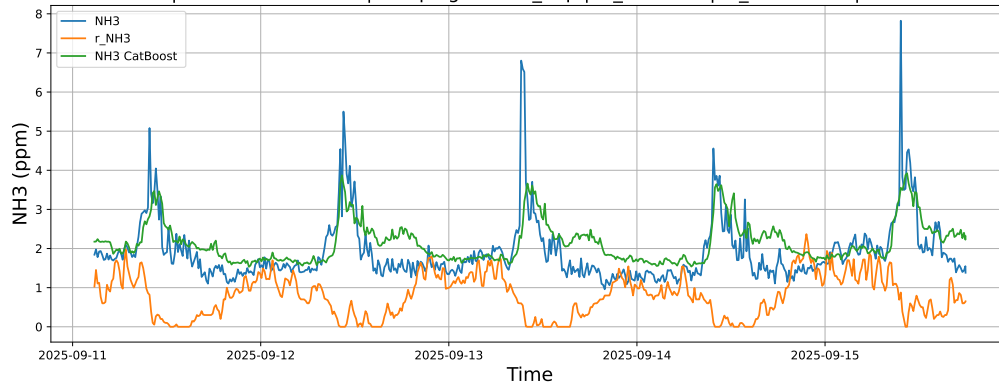
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



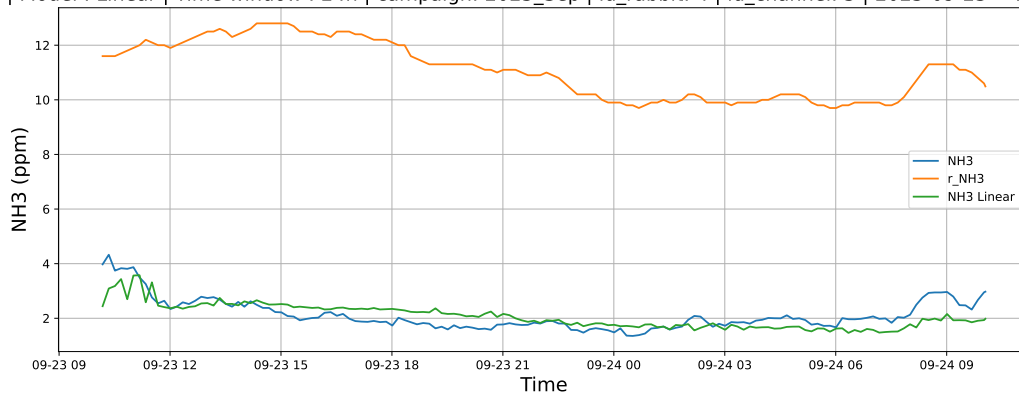
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



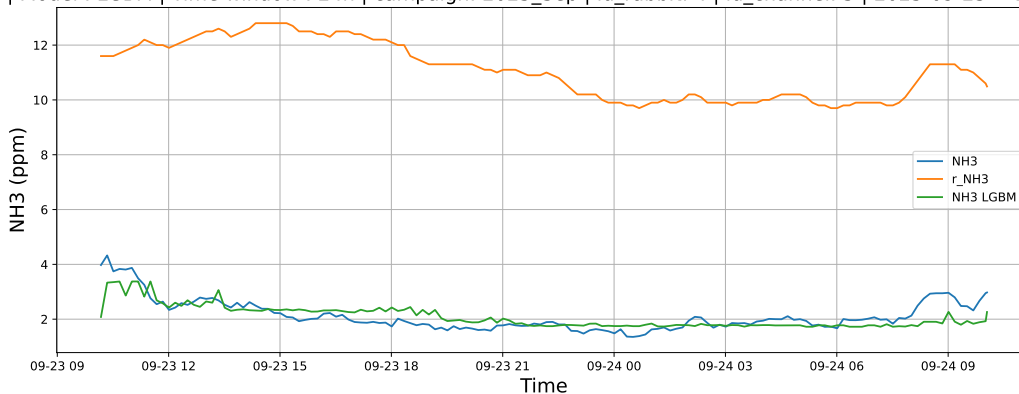
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

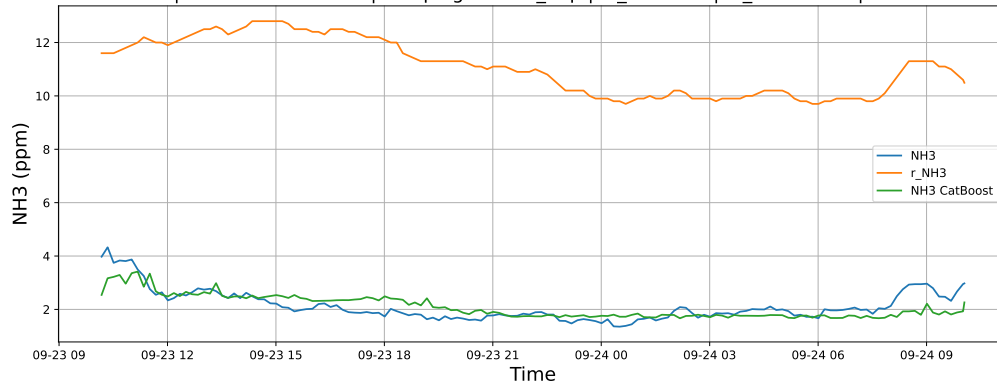
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

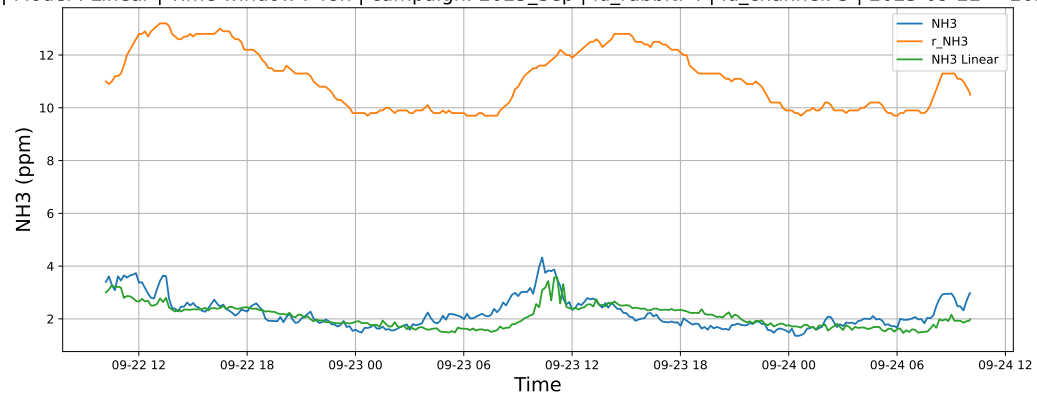


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

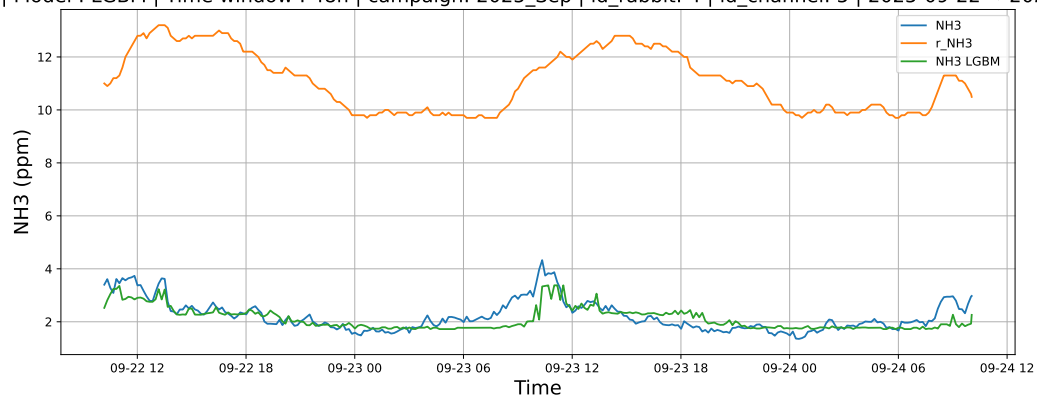


4.1.7.2 Time window : 48

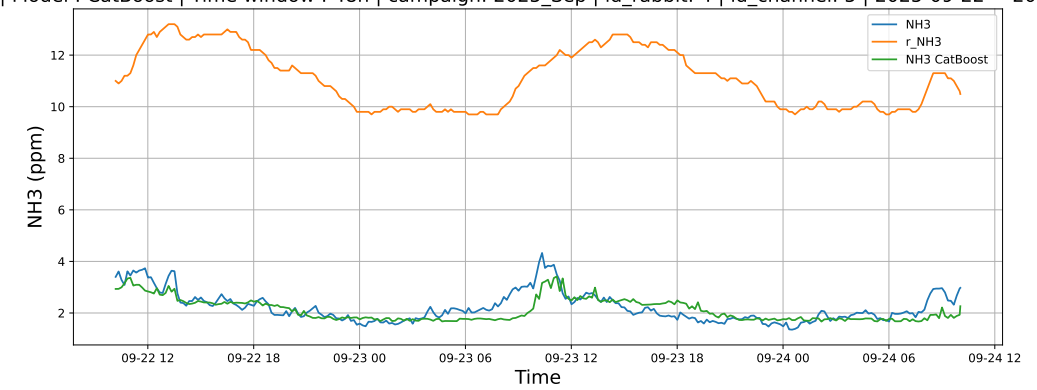
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

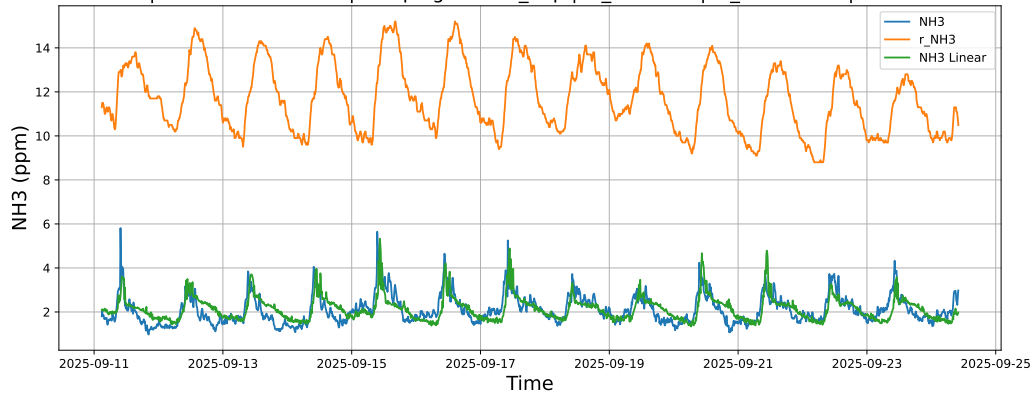


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

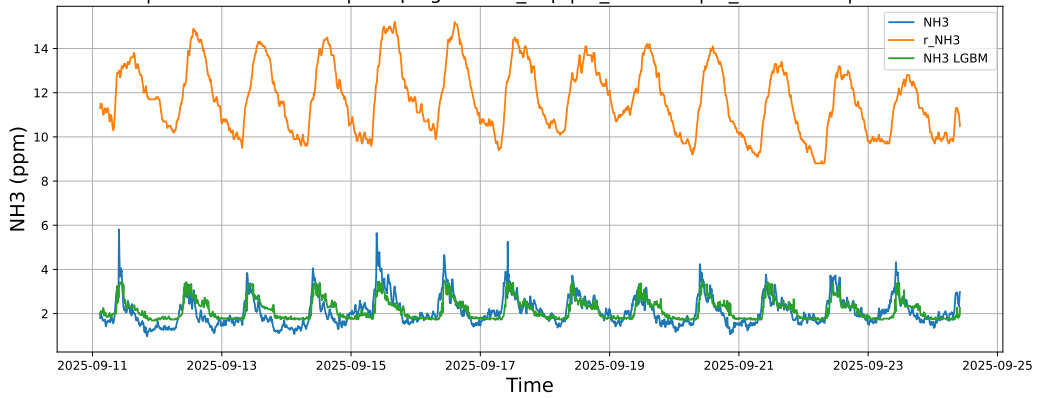


4.1.7.3 Time window : ALL

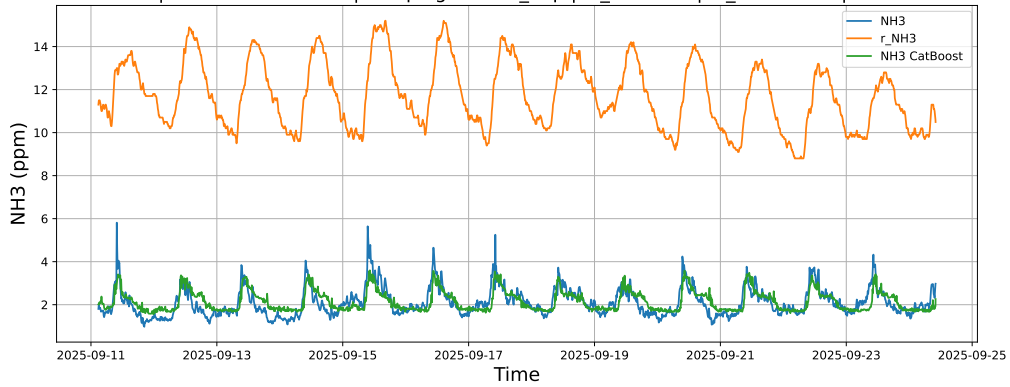
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



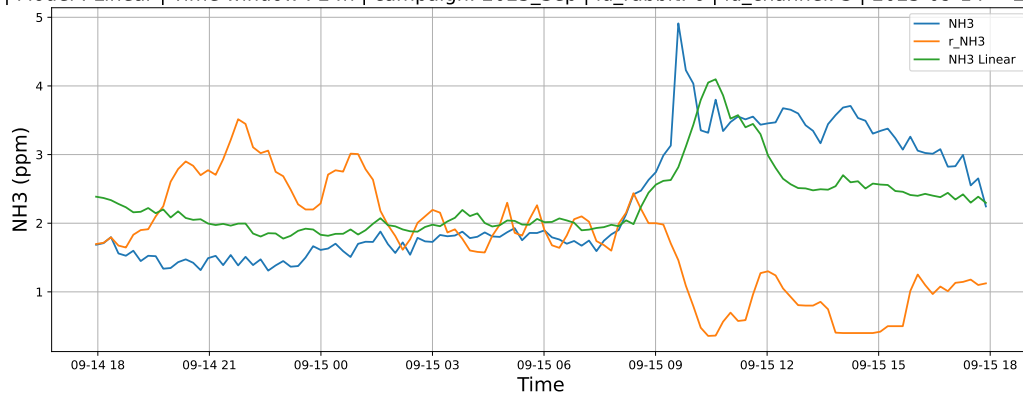
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



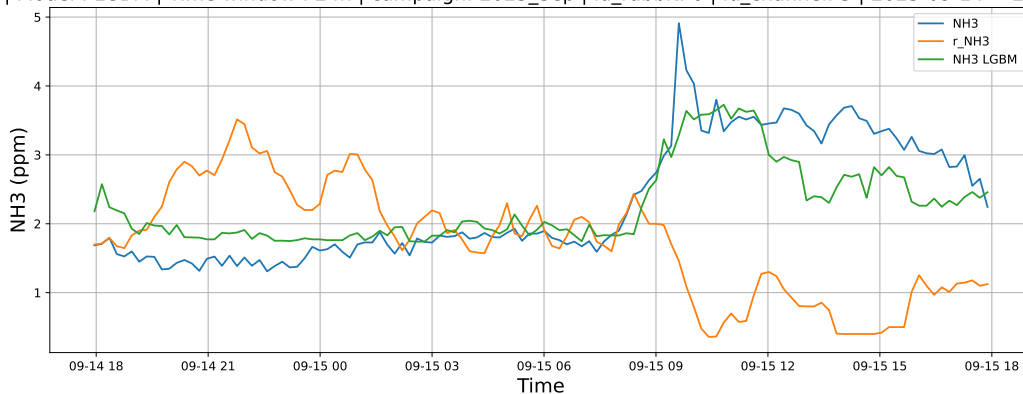
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

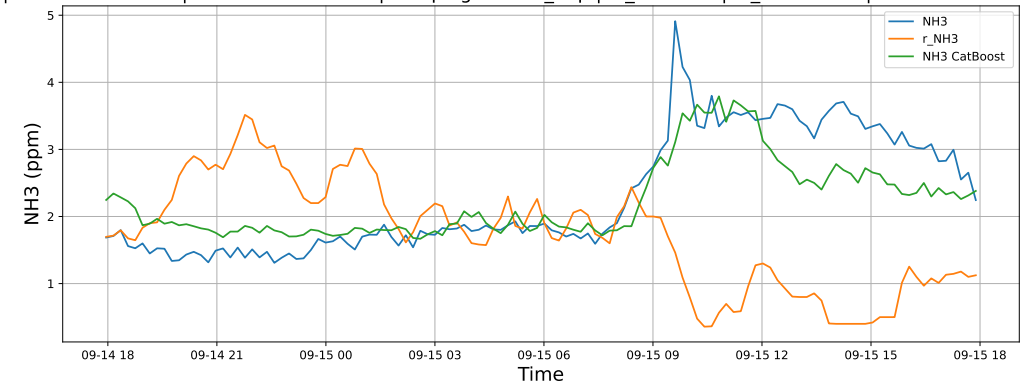
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

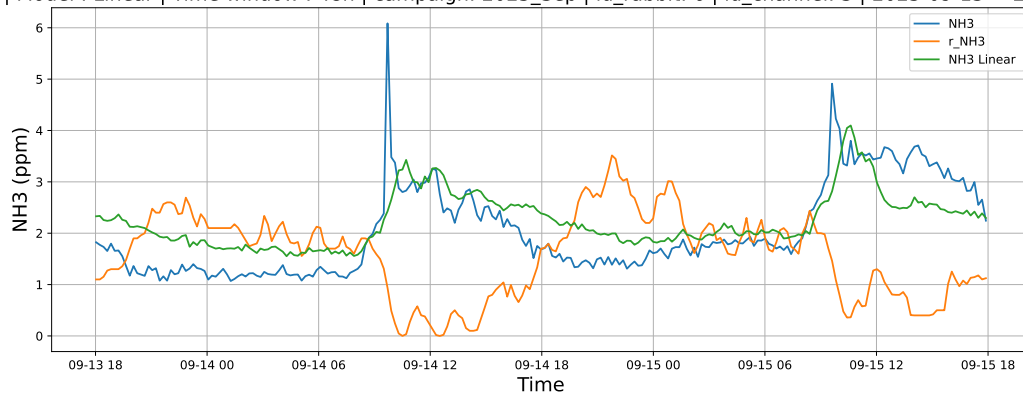


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

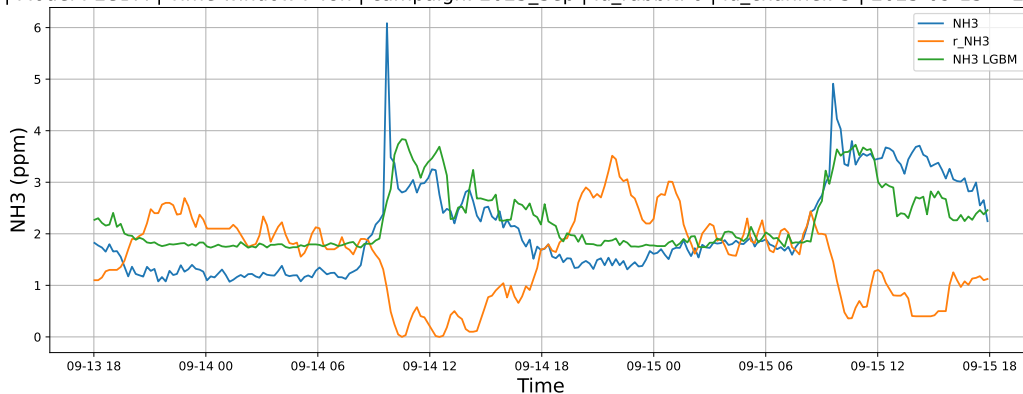


4.1.8.2 Time window : 48

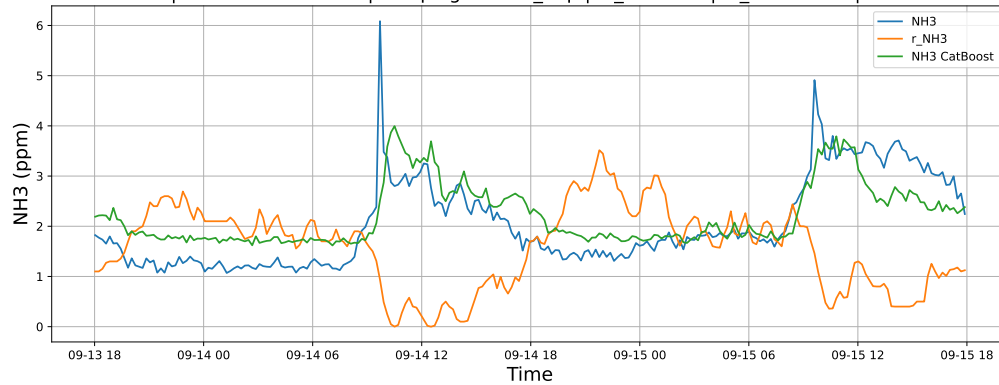
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

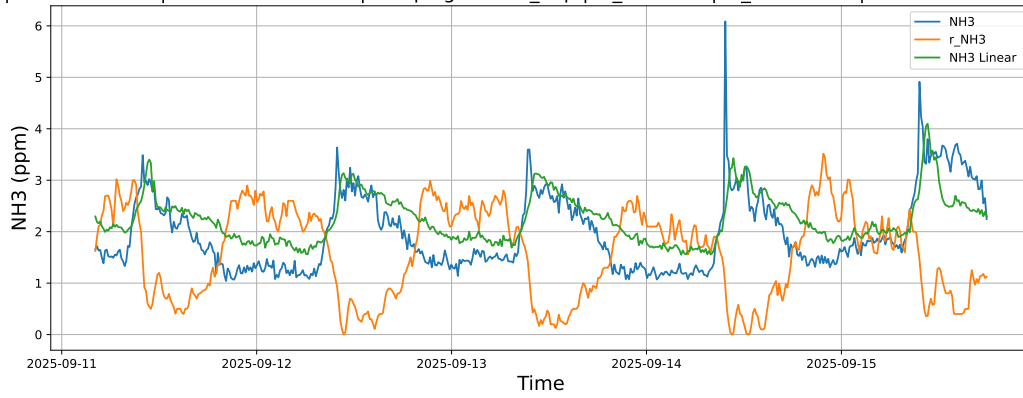


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

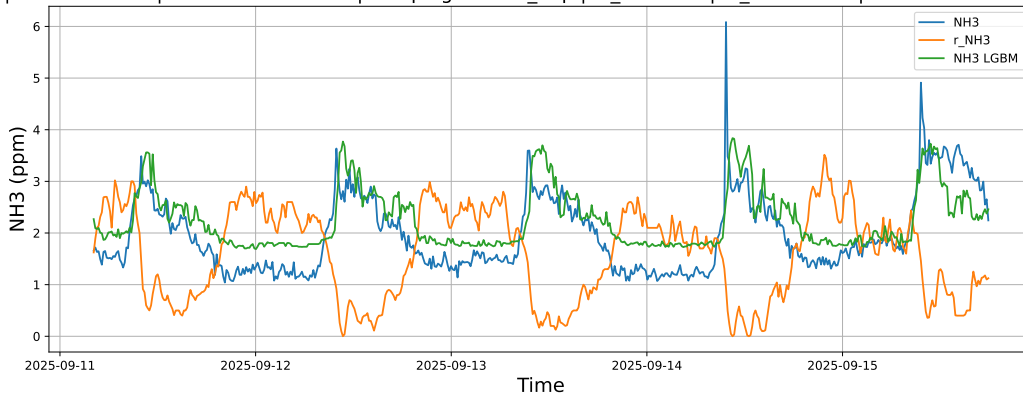


4.1.8.3 Time window : ALL

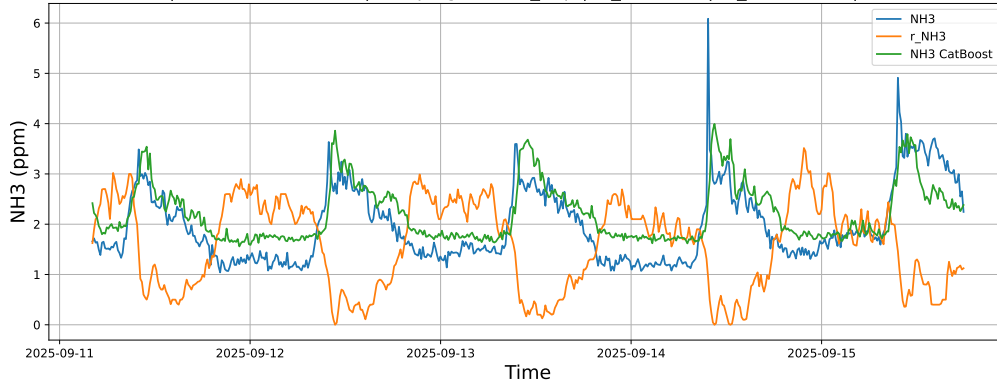
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

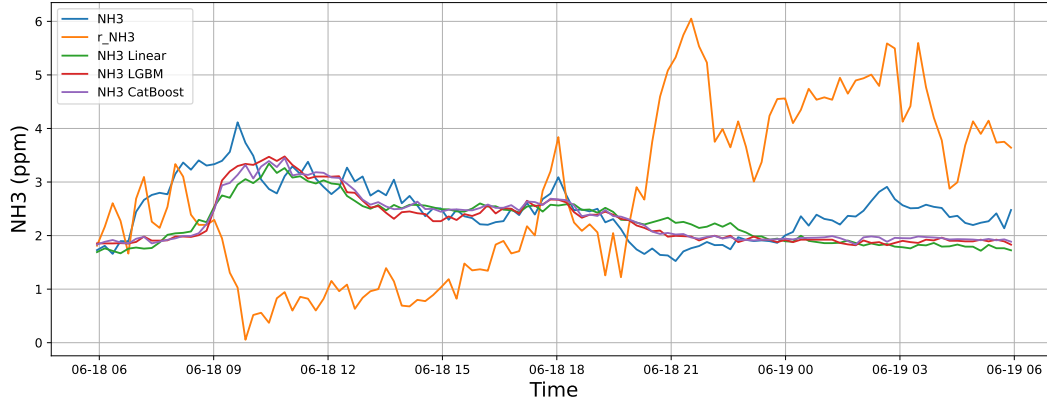


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

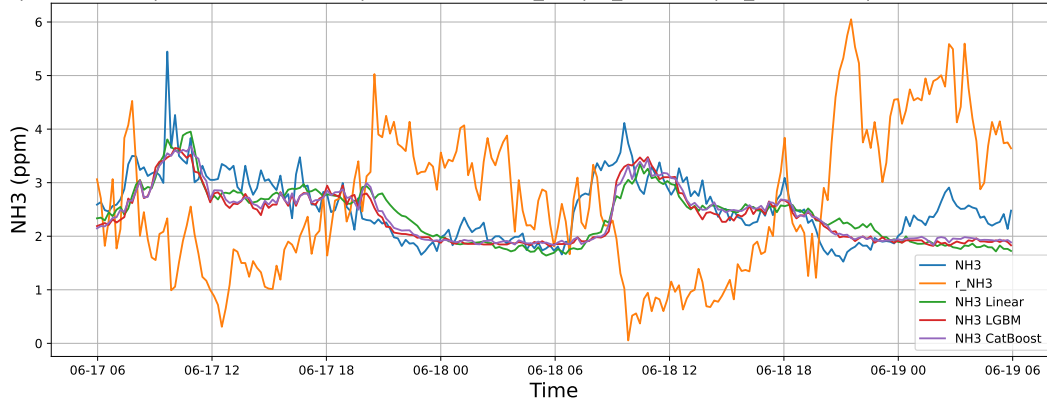
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



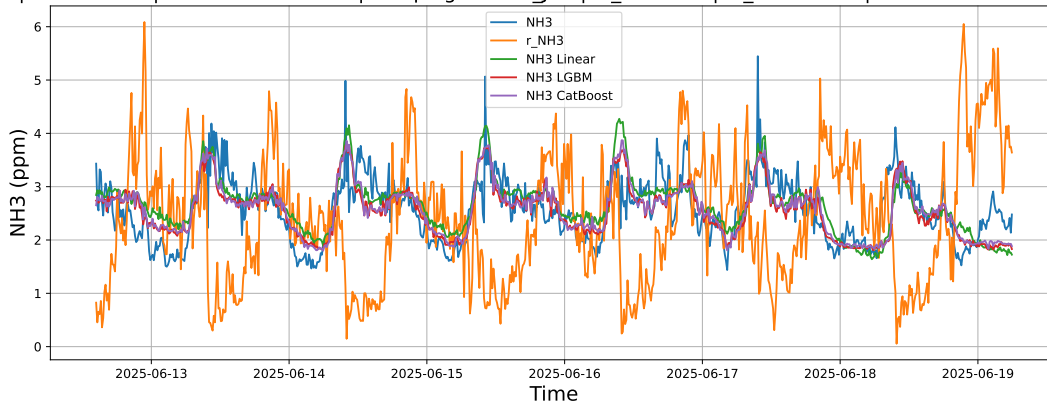
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

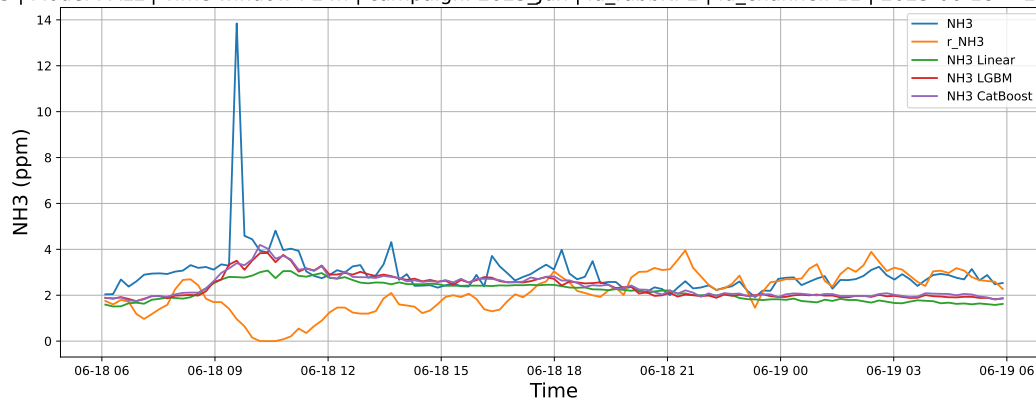
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

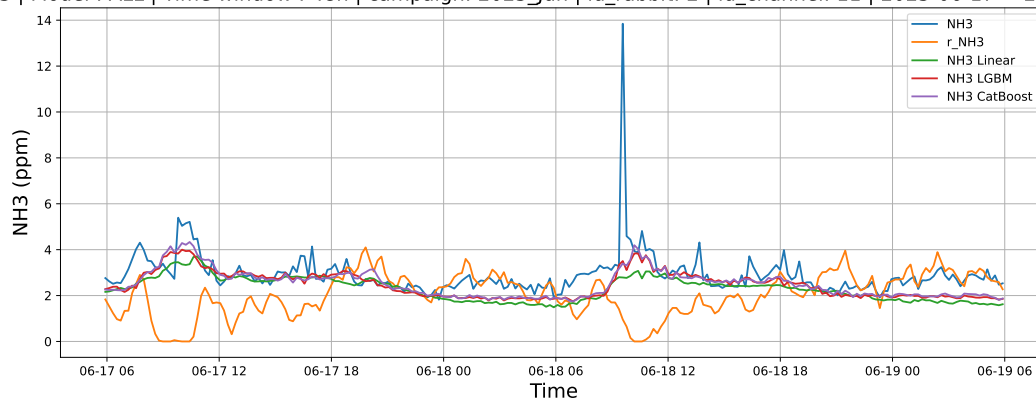
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



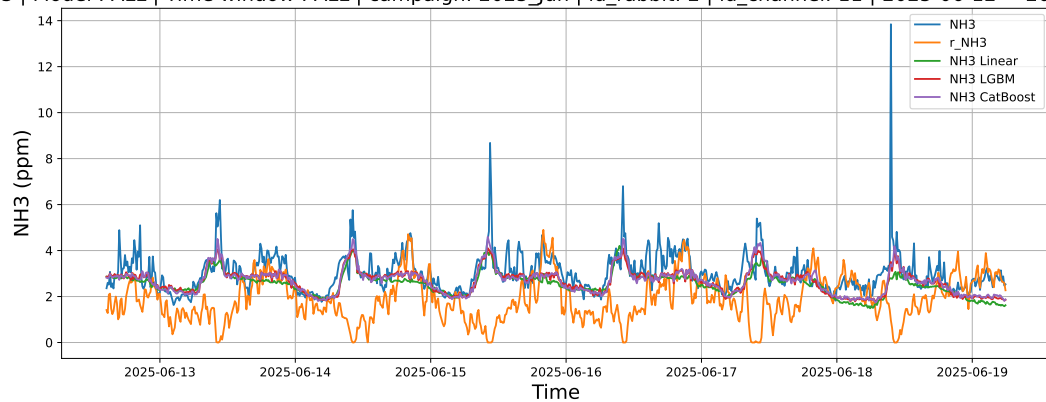
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

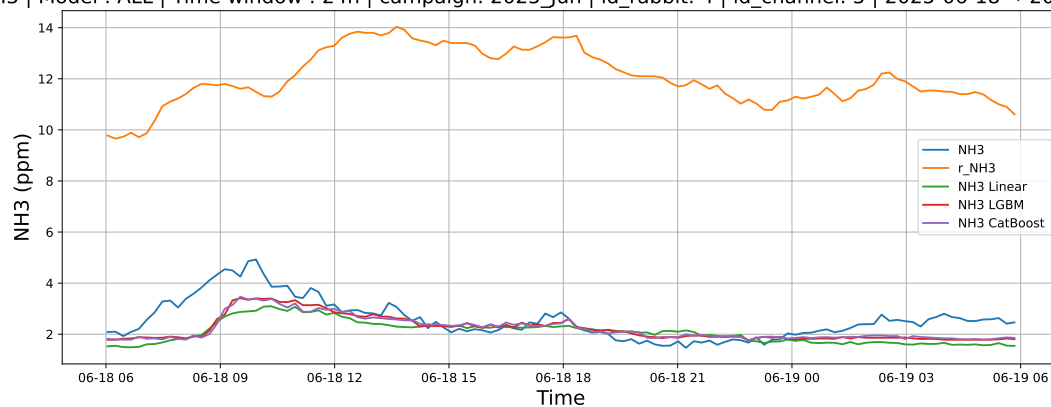
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

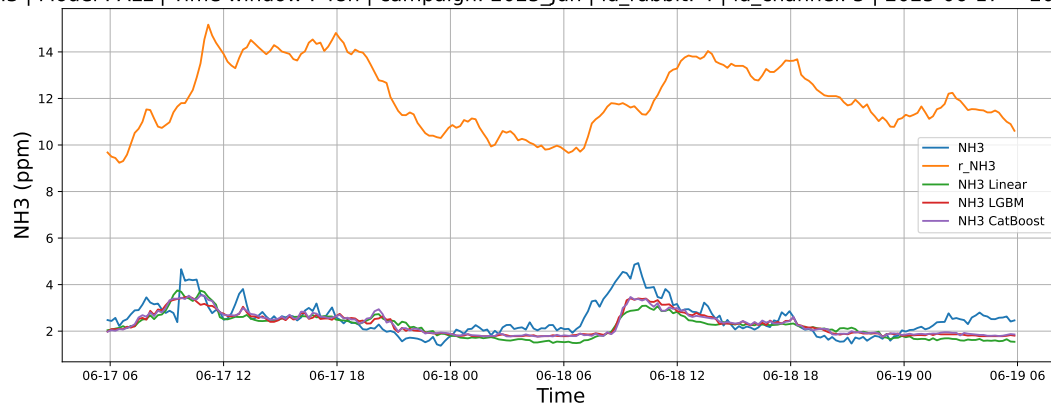
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



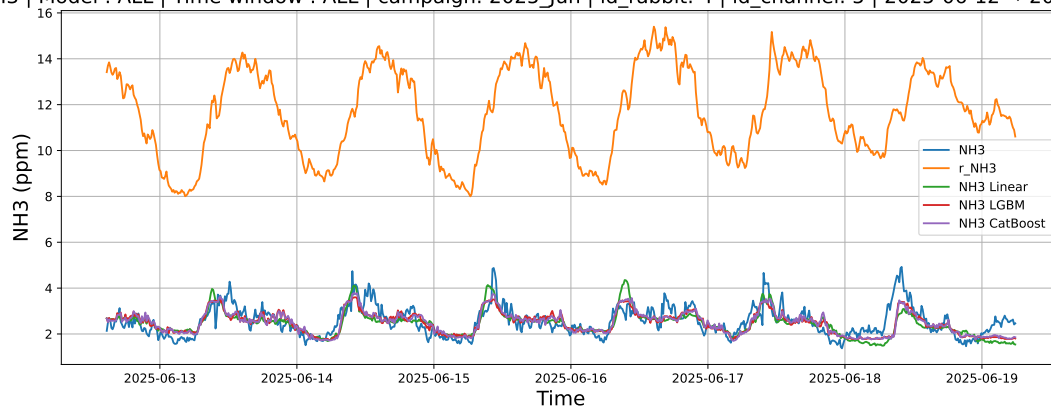
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

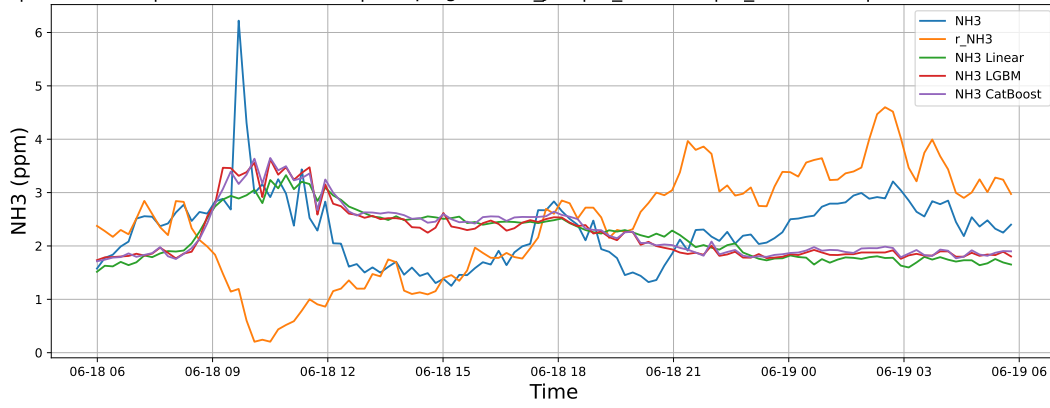
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

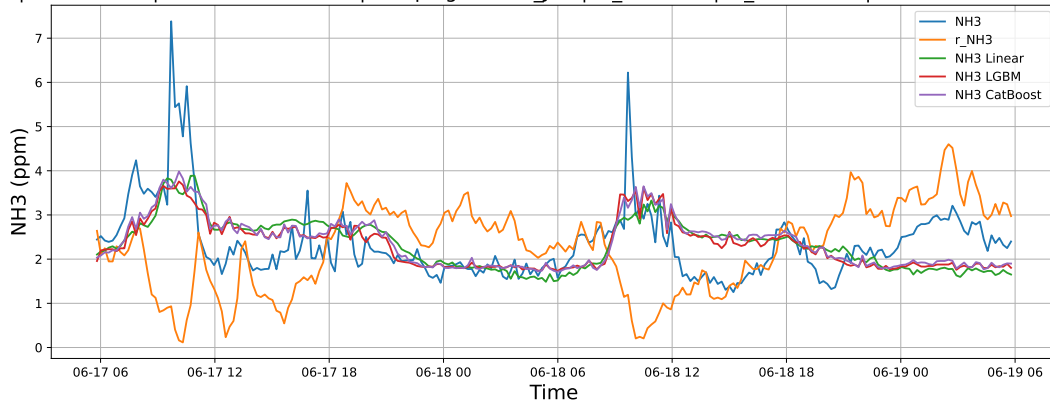
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



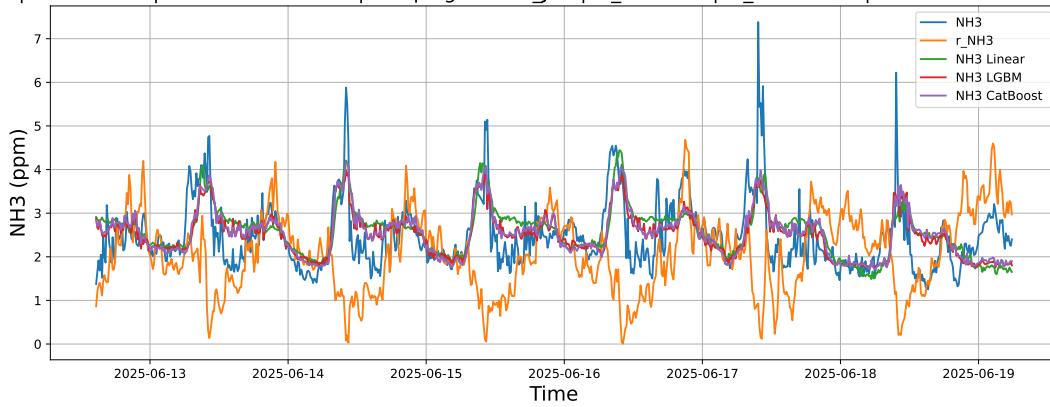
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

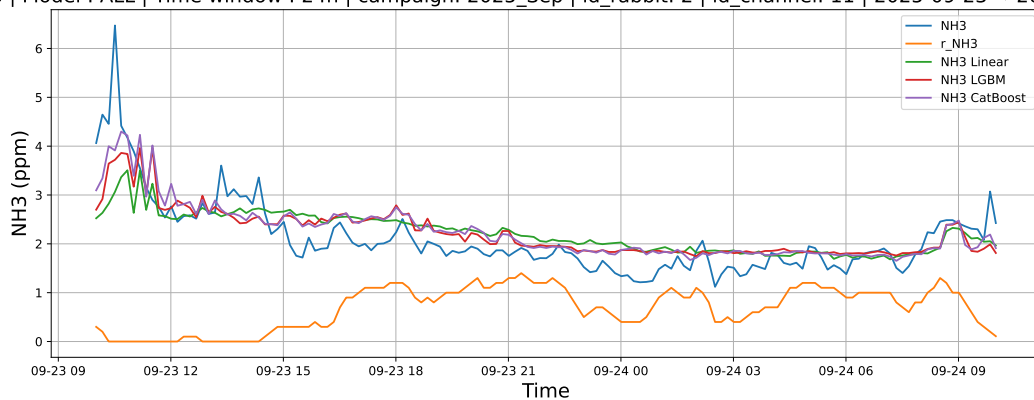
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

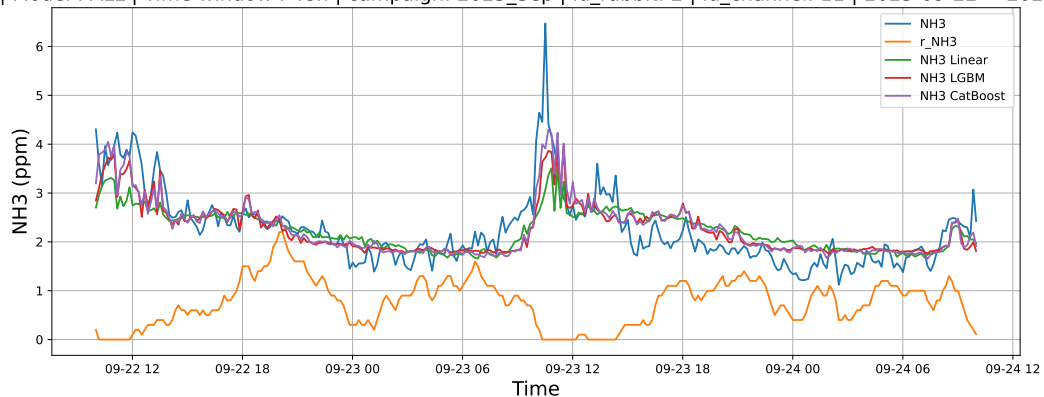
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



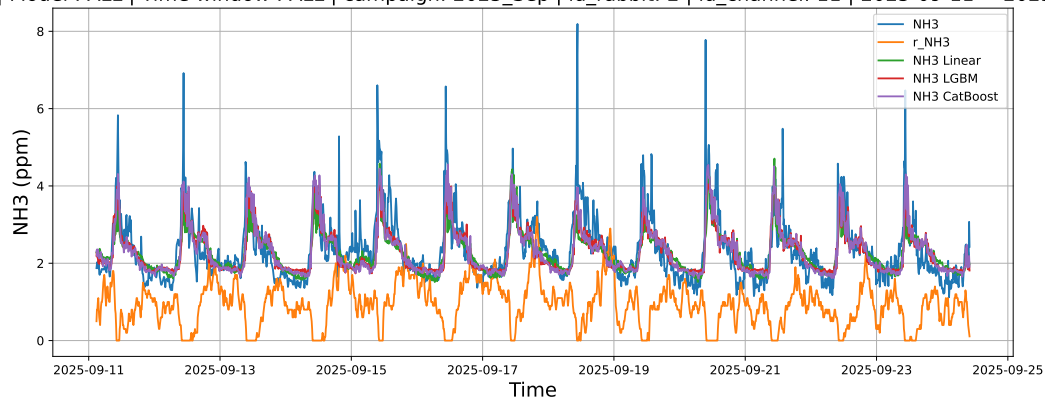
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

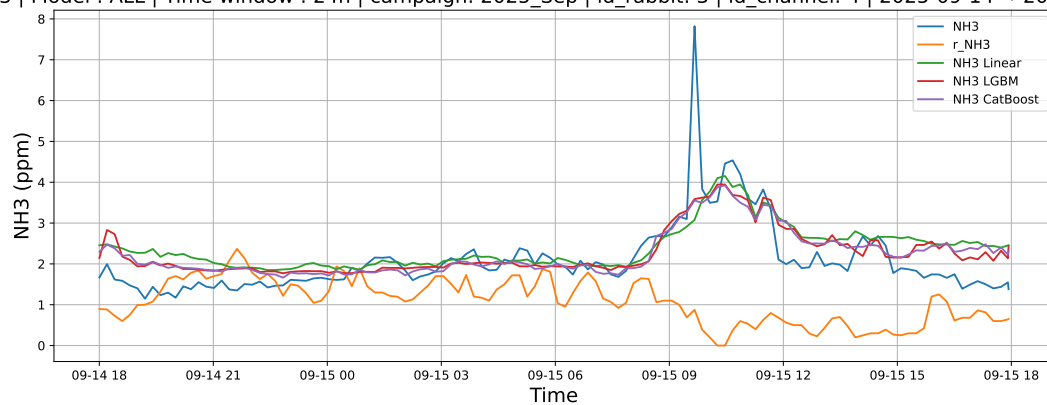
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

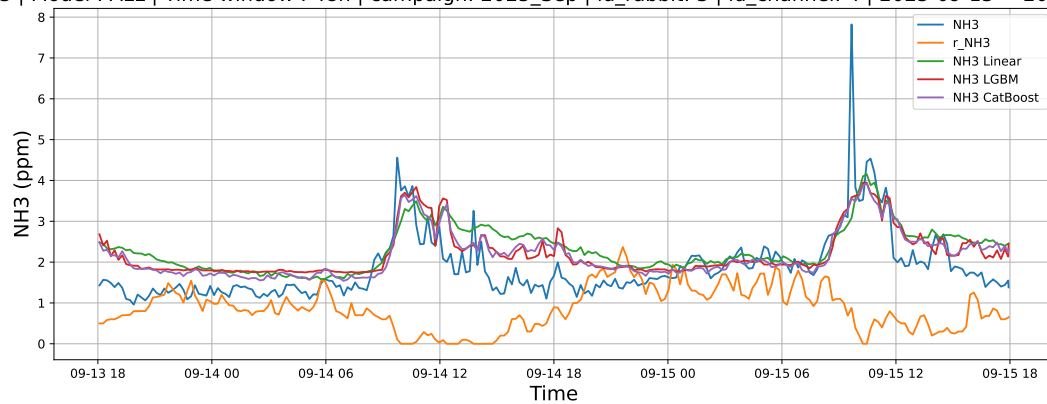
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



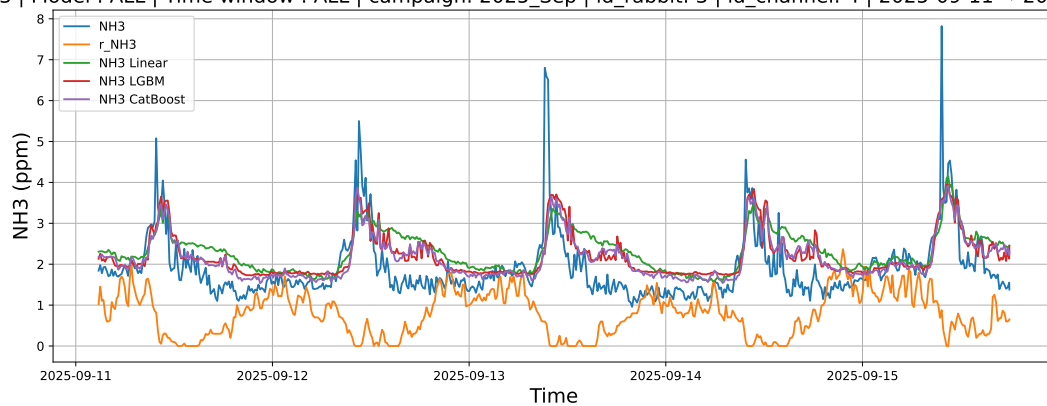
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

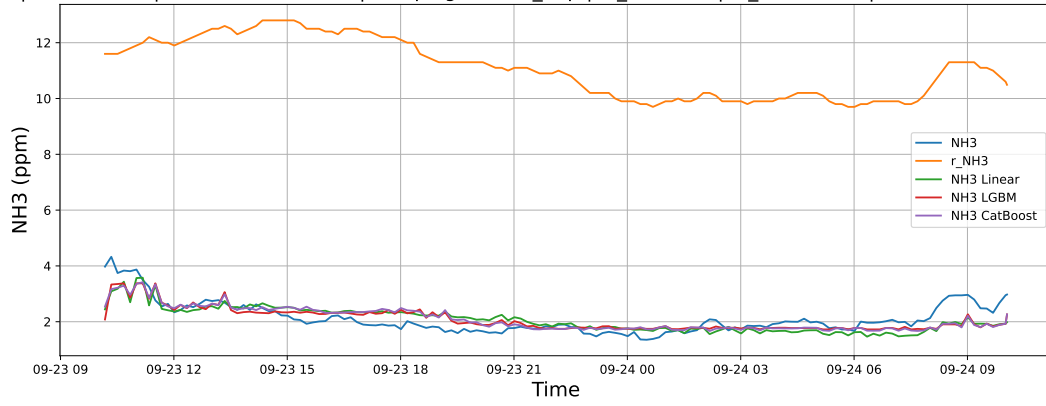
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

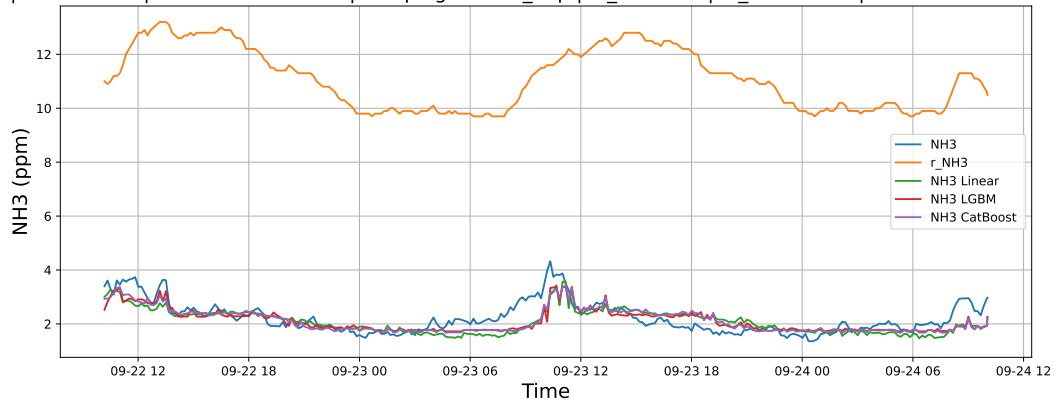
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



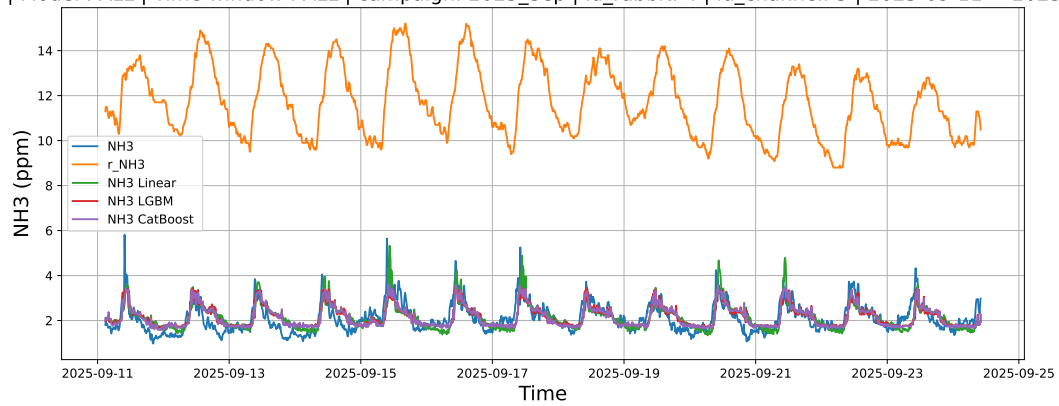
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

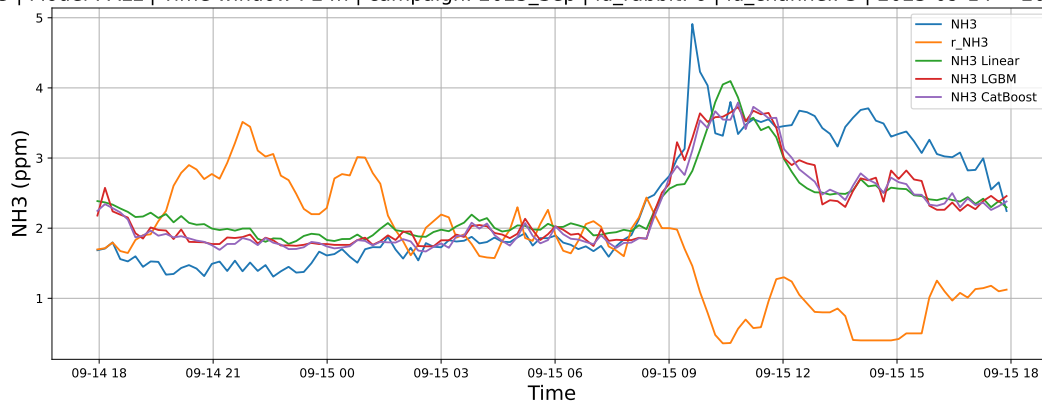
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

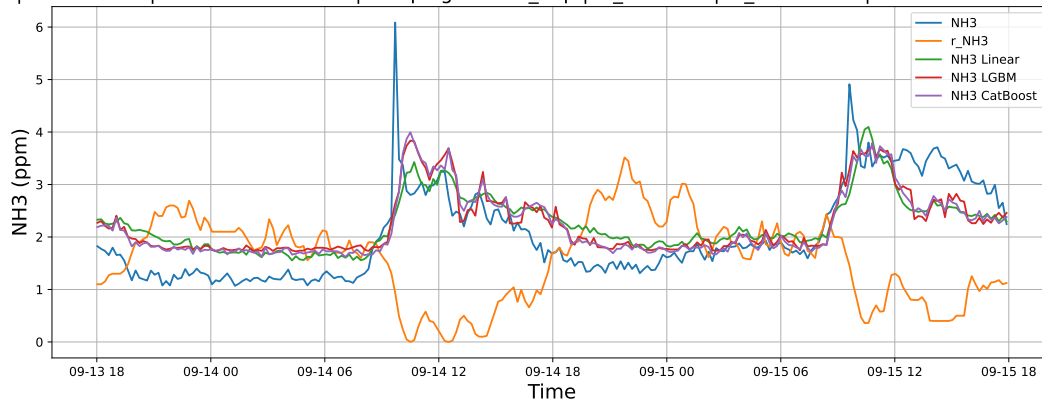
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

